

Adventist University of Central Africa

AGUKA SMART FARMING KIT

Imbaraga Farmers Organization

A final year project presented in partial fulfilment of the requirements for the
degree of BACHELOR OF SCIENCE IN INFORMATION TECHNOLOGY

Major in
Information Management

By Adolphe Uwayo

September 2025

ABSTRACT

Adventist University of Central Africa

Project research of the Bachelor's degree in Information Technology.

Major of Information Management.

TITLE: AGUKA SMART FARMING KIT

RESEARCHER: UWAYO Adolphe

SUPERVISOR: Eng. NSENGIYUMVA Juvenal

Date Completed:

Farming is still the most important part of the Rwandan economy because it gives most of the jobs in rural areas. Even though farming is important, most small-scale farmers still use the old-fashioned way, which is mostly based on guesswork and trying things out. It makes them more likely to have problems with climate change, soil erosion, and not using resources wisely. Their lives are also hard because there aren't many new technologies that work well.

Therefore, the use of the AGUKA SMART FARMING KIT is suggested as a new technology that can help increase food production by assisting farmers to track the conditions within the farm and make decisions concerning agriculture. The technology is architected to integrate with physical Internet of Things sensors that collect information about moisture, temperature, humidity, and light within the farm. For this prototype implementation, sensor data is generated through a validated software simulation module that follows Rwanda's seasonal agricultural patterns, a methodology consistent with smart farming system prototyping. Data is transmitted through a mobile application, web platform, and USSD interface.

The main objective of this project is to develop and implement "AGUKA SMART FARMING KIT" For "Imbaraga Farmers Organization." By providing useful information that supports effective irrigation techniques, crop choices, and higher farm output, this will aid farmers in closing the digital gap in Rwanda's agriculture sector. The technology stack includes IoT simulation, React.js, Flutter, Node.js, Express.js, PostgreSQL, and Firebase Cloud Messaging.

DECLARATION

I **Adolphe UWAYO**, student at Adventist University of Central Africa in Faculty of Information Technology, Department of Information Management with Registration N° 25743, I do hereby declare that this research project report is entirely the real reflection of my own original work and experience to the best of my knowledge. It has never been either partially or wholly presented in any university or any higher learning institution.

Signature.....

Date: .../...../.....

APPROVAL

I, **Juvenal NSENGIYUMVA**, hereby certify that this project report, has been done under my supervision and submitted with my approval.

Signature:

Date: .../...../.....

DEDICATION

With immense joy, I dedicate this Research Project

To God almighty,

To my dear parents,

To all my family members,

To all my friends and colleagues,

To my guide for his valuable guidance.

Moreover, to all my classmates at AUCA.

TABLE OF CONTENTS

ABSTRACT	i
DECLARATION	ii
APPROVAL	iii
DEDICATION	iv
LIST OF TABLES	ix
LIST OF FIGURES	xi
LIST OF ABBREVIATIONS	xii
ACKNOWLEDGEMENTS	xiii
CHAPTER 1	1
GENERAL INTRODUCTION	1
Introduction	1
Background of the Study	2
Statement of Problem	3
Choice and Motivation of the study	4
Objectives of the study	5
General Objective	5
Specific Objectives	5
Scope of Study	6
Methodology and Techniques used in the study	7
Interview	7
Documentation	7
Research questions	7
Expected Results	8

Organization of the Report.....	9
CHAPTER 2	10
ANALYSIS OF EXISTING SYSTEM	10
Introduction	10
Description of Current System Environment.....	11
Historical Background.....	11
Vision.....	11
Mission	11
Description of Current System.....	12
Analysis of the Current System.....	13
Modeling of The Current System	14
Problems of the Current System.....	15
Proposed Solutions.....	17
System Requirements.....	18
Functional Requirements.....	18
Non-Functional Requirements.....	19
CHAPTER 3	21
REQUIREMENTS ANALYSIS AND DESIGN OF THE NEW SYSTEM.....	21
Introduction	21
Unified Modeling Language (UML).....	22
UML models.....	23
Design of the new system.....	29
Class Diagram.....	43
Sequence Diagram	44

The Elements of a sequence diagram	45
Activity Diagram	53
Database Diagrams	55
Data Dictionary.....	56
System Architecture Design	65
CHAPTER 4	66
IMPLEMENTATION OF THE NEW SYSTEM.....	66
Introduction	66
Technologies used	67
Frontend.....	67
Backend	68
IoT Sensor Simulation	70
Presentation of the New System	71
Software Testing	83
Hardware and Software Compatibility Requirements	85
CHAPTER 5	88
CONCLUSIONS AND RECOMMENDATIONS	88
Conclusion.....	88
Recommendations	89
REFERENCES	91
Books.....	91
Journals.....	91
Websites	91
APPENDICES	93

Curriculum Vitae..... 94

Data Collection Letter 95

Approval Letter from Organization..... 96

LIST OF TABLES

Table 1: use case for login	32
Table 2: use case for manage farm profile.....	33
Table 3: use case for view soil & weather data	34
Table 4: use case for manage users & access control	35
Table 5: use case for manage irrigation recommendations.....	35
Table 6: use case for record farm activities	36
Table 7: use case for monitor assigned farms	37
Table 8: use case for generate cooperative-level report.....	38
Table 9: use case for view audit logs	38
Table 10: use case for push notification	39
Table 11: use case for view recommendations & insights.....	40
Table 12: use case for view performance-based recommendations.....	41
Table 13: use case for view analysis results	43
Table 14: Sequence diagram elements.....	46
Table 15: Users Table Data Dictionary	57
Table 16: Farmer Profile Table data dictionary	58
Table 17: Crop Table Data dictionary	58
Table 18: FarmerCrop Table data dictionary.....	59
Table 19: Sensor Table Data dictionary.....	59
Table 20: SoilReading Table Data Dictionary.....	60
Table 21: WeatherReading Table Data Dictionary.....	61
Table 22: IrrigationSchedule Table Data Dictionary.....	61
Table 23: Alert Table Data Dictionary	62

Table 24: Advisory Report Data Dictionary	62
Table 25: CooperativeReport Table Data Dictionary	63
Table 26: System Report Table Data Dictionary	64

LIST OF FIGURES

Figure 1: The current system modeling	14
Figure 2: administration use case diagram.....	29
Figure 3: operational users use case diagram	30
Figure 4: class diagram	44
Figure 5: User Login Sequence Diagram.....	47
Figure 6:Manager Farm Profile Sequence Diagram	48
Figure 7: Irrigation Recommendation Sequence Diagram	50
Figure 8: View Soil & Weather Data Sequence Diagram	51
Figure 9: Generate Cooperative reports Diagram.....	52
Figure 10: Administration Activity Diagram.....	53
Figure 11: Activity Diagram for User operations	55
Figure 12: Diagram for Database Schema	56
Figure 13: System Architecture Design.....	65
Figure 14: Welcome Page.....	71
Figure 15: Sign up page	73
Figure 16: Login page.....	74
Figure 17: Farmer dashboard.....	75
Figure 18: Field Officer Dashboard.....	76
Figure 19: Cooperative Manager Dashboard.....	77
Figure 20: Admin Overview Dashboard.....	78
Figure 21: Super Admin Console Overview.....	79
Figure 22: Super Admin Report.....	80
Figure 24: PDF Report.....	82

LIST OF ABBREVIATIONS

AUCA	Adventist University of Central Africa
CSS	Cascading Style Sheet
DBMS	Database Management System
FK	Foreign Key
GB	Giga Byte
HTML	Hyper Text Markup Language
ICT	Information and Communications Technology
IDE	Integrated Development Environment
IT	Information Technology
IoT	Internet of Things
MB	Mega Byte
OOA	Object Oriented Analysis
OOD	Object Oriented Design
OOM	Object Oriented Methodology
PHP	Hypertext Preprocessor
PK	Primary Key
SRS	Software Requirement Specifications
UML	Unified Modeling Language
UP XML	Unified Process - extensible Markup Language

ACKNOWLEDGEMENTS

First and foremost, I would like to give all credit and gratitude to God for the gift of life he bestowed on me, and for the wisdom and understanding which he gave me so that my existence can be justified.

In all sincerity, I want to thank the academic staff members of the faculty of Information Technology and all the rest administration members of the Adventist University of Central Africa for the hard work they have done while I studied in this university. The knowledge and wisdom which I received helped me in doing this work, and now I take this opportunity to thank all the staff members of AUCA.

The special appreciation goes to my supervisor, Eng. NSENGIYUMVA Juvenal, who, despite the numerous

activities he has to do in a day and a great number of his obligations, agreed to be my supervisor as much as he could. His advice, guidance, insight, patience and help were really important for completion of the work. For his professional guidance and insight which helped me on my long path to finishing the research work, I am thankful.

I am also thankful to my parents, my brothers and sisters and all my family members who encouraged me in this journey.

I am deeply grateful to the management and farmers from Imbaraga Farmers Organization for their willingness to participate in gathering data and allow me access to the working environment of their organization. Their ideas played a crucial part in designing the proposed system.

In the course of this distant education at AUCA, I was unable to cope with everything by myself. My colleagues have assisted me in many aspects starting from the very first moments. Anyone who helped me in any way during our joint work, I sincerely thank you; your contribution allowed me to finish the task as well as possible.

May God bless you!

UWAYO Adolphe

CHAPTER 1

GENERAL INTRODUCTION

Introduction

In the last three decades, a number of achievements have been realized in Rwanda as evidenced by fast economic growth and significant declines in poverty rates. The agricultural industry has proved to be a key sector that has played an integral role in improving the wellbeing of citizens and driving the national economy.

Currently, the agricultural industry makes up for 25% of the nation's GDP. As important as the agricultural industry may be from an economic standpoint, many farmers lack modern technologies and information on farming. With climate change, degradation of soils, and increased population, traditional ways of soil testing, water management, and crop management prove to be ineffective.

ICT has become a very strong tool for bringing about an agricultural revolution around the world. Precision farming is also known as smart farming and it utilizes ICTs such as IoT, data analysis, artificial intelligence, and mobile computing for decision-making and efficient utilization of resources in farming activities. It allows farmers to monitor field conditions, be informed about relevant situations, and take data-backed decisions which increase their crop yield.

Aguka Smart Farming Kit refers to the locally modified smart farming technology which is tailored specifically for the needs of the smallholder farmers of Rwanda. It has been specifically developed with the capability to send and communicate environmental information as well as soil information from the farms to the farmers themselves through digital means of communication such as smartphones, websites, and USSD.

This report contains information about the complete procedure used in researching, designing, and developing the Aguka Smart Farming Kit. This information is based on academic research and real-life data gathered from farmers involved with the Imbaraga Farmers Organization in Rwanda.

Background of the Study

Agriculture is one of the major industries of the economy of Rwanda that contributes to food security in the country. Many farmers use it to earn their living in the country. However, even though the importance of this industry is high, there are still various factors influencing the efficiency of smallholders. For example, according to the information from the Rwanda Agriculture Board, the vast majority of farmers use small lands – under two hectares each – and rely on irrigated farming that requires a lot of rains.

The unpredictability of the weather in the region along with various other factors is a source of additional challenges for the agriculture of the country that affects its productivity level negatively. Those factors include soil degradation due to constant cultivation without any restoration, regular diseases of plants and pests, and lack of affordable fertilizers among others. Due to the aforementioned difficulties, the productivity rate of farmers is low and it is challenging to maintain sustainable development of agriculture.

There are digital technologies in the field of agriculture that are called smart farming or digital agriculture. The mentioned innovations are used by means of mobile applications, sensors, data analyses, and decision-making systems to help farmers make right decisions. There are various initiatives of development organizations and technology innovators that introduce solutions of smart agriculture to increase efficiency, yields, and decrease risks of farmers. However, those technologies have proved to be quite useful but there is not enough number of farms where they are adopted.

Nevertheless, some of the current solutions of smart agriculture are not completely suitable for the specific local environment in Rwanda. Some innovations may be expensive or difficult to understand or use. Moreover, certain technologies cannot be easily used because they need additional internet access or have other requirements that are hard to meet in the area. This study will focus on the problem of lack of proper technology innovation in agriculture.

For this research, the Imbaraga Farmers Organization will be chosen as the case study. It was created to help smallholders with training and resources to farm. The environment where this organization operates, the nature of its members, and the current state of its data management provide a realistic basis for the identification of the system requirements.

Statement of Problem

Smallholder farmers in Rwanda have long struggled with challenges that affect their productivity, income, and ability to sustain their farms. A key problem is that agricultural extension services often miss the people who need them most, leaving many farmers without timely advice or support.

The effectiveness of training programs provided to farmers cannot be properly measured. This makes it difficult to evaluate their impact or improve future interventions. Collective activities such as purchasing inputs and marketing agricultural products also lack a reliable data foundation. As a result, poor decision-making and missed economic opportunities occur.

Currently, most farming data is managed using a paper-based system, where agronomists record information manually using pens and notebooks. This handwritten approach is time-consuming, prone to errors, and makes data storage, retrieval, and analysis very difficult.

ICT is an important element that assists in reducing the complexities associated with making tasks easy for human beings while increasing accuracy and minimizing errors. One such device that has been created to assist with this is the Aguka Smart Farming Kit. This is a technological innovation aimed at making management and conducting of agriculture related functions easier.

The identified challenges demonstrate that the current farming information management approach is inefficient, slow, and incapable of supporting modern precision agriculture practices. Without a centralized digital platform, farmers continue to experience delayed decision-making, limited access to critical information, poor data management, and reduced agricultural productivity. Therefore, there is a need for an integrated smart farming solution capable of collecting, processing, and analyzing agricultural data in real time.

Choice and Motivation of the study

This project is motivated by the need to bridge the digital gap in Rwanda's agricultural sector. As an Information Management student at AUCA, contributing a practical ICT solution for smallholder farmers represents both academic fulfilment and a direct contribution to national development priorities aligned with Rwanda's Vision 2050. The system is intended to support improved agricultural activities by providing accurate and real-time information such as weather updates, soil conditions, and best farming practices. Through this, farmers can enhance decision-making, increase productivity, and promote more sustainable agricultural operations.

In addition, the study is driven by the increasing importance of integrating technology into agriculture to address challenges faced by smallholder farmers, including unpredictable weather patterns, limited access to timely information, and inefficient resource utilization. By introducing an IoT-based smart farming system, the project aims to improve efficiency and reduce reliance on traditional farming methods that often lead to low yields and high operational risks. The system will also help farmers reduce production costs by optimizing the use of inputs such as water, fertilizers, and labour through data-driven recommendations. It will further support early detection of farm-related issues, allowing timely intervention before significant losses occur.

Furthermore, from an academic perspective, this research fulfills the requirements for obtaining a Bachelor's degree at the Adventist University of Central Africa (AUCA). It also provides an opportunity to apply theoretical knowledge gained throughout the program into a real-world solution, strengthening both technical and problem-solving skills while contributing to meaningful societal impact. The project also enhances research and development skills by involving system analysis, design, and implementation of IoT-based solutions. It encourages innovation and critical thinking in addressing real agricultural challenges using technology.

Lastly, the study is expected to encourage digital adoption among farmers by demonstrating the practical benefits of smart farming technologies. This can help promote long-term behavioral change in agricultural practices, leading to a more technology-driven and resilient farming community. It will also contribute to improving collaboration between farmers and agricultural experts through better communication channels. Ultimately, the system aims to support food security and sustainable agricultural growth in Rwanda.

Objectives of the study

In regard to my research, there are two types of objectives which can be considered. These are the general and specific objectives as mentioned below.

General Objective

The primary objective of this study and research is to develop a system that will allow farmers to optimize their agricultural operations in a better way and will help them become more digitized through the use of the system in form of a web application and mobile app.

Specific Objectives

The objectives of the research include the following in pursuit of the primary objective:

- To develop a computerized system for managing the data of the farmers.
- To facilitate real-time information such as weather forecast and the soil condition.
- To facilitate better communication between the farmers and the agronomists.
- To eliminate the shortcomings of the current method of using paper.
- To aid in effective decision making in relation to agriculture.
- To integrate a rule-based AI recommendation engine that analyses simulated soil conditions, weather forecasts, and crop data to generate personalized farming recommendations for farmers, extension officers, and cooperative managers.

Scope of Study

This research project will focus on the development and implementation of Aguka Smart Farming Kit as a digital intervention solution using IoT to support smallholder farmers from Imbaraga Farmers Association in Rwanda. The proposed system will rely on IoT devices that will continuously monitor various farm-related parameters such as moisture, air temperature, air humidity, and light intensity. The system will collect the data and process it to be disseminated using web-based and mobile applications by both farmers and agronomists. The proposed solution will leverage ICT and IoT technologies in order to enhance data-driven decisions, improve farm productivity, and promote efficient and sustainable farming practices.

The scope of this study also includes the design and development of a centralized data management system that enables efficient storage, processing, and retrieval of farm data collected from different sensors. This system will ensure that all environmental and agricultural data is properly organized and made accessible to authorized users in real time. It will also support the integration of analytical tools that help in generating meaningful insights from raw sensor data.

In addition, the study covers the development of user-friendly interfaces for both mobile and web platforms to ensure ease of access for farmers and agricultural officers. These interfaces will allow users to monitor farm conditions, receive recommendations, and access alerts or notifications regarding farm activities. The system will be designed to support different user roles with specific functionalities tailored to their responsibilities within the farming ecosystem.

Furthermore, the research is limited to smallholder farmers within the Imbaraga Farmers Association in Rwanda, focusing on improving their farming practices through digital transformation. It does not extend to large-scale commercial farming systems or other agricultural regions outside the targeted scope. The study emphasizes practical implementation, usability, and real-world application of IoT-based smart farming technologies to improve agricultural productivity and sustainability.

Methodology and Techniques used in the study

This subtopic describes the difference facts and finding techniques that were used for achieving the goals and objectives of the project such as Population of the study, Data collection and Analysis, system analysis, system design and implementation, Testing and validation.

Interview

Interview was the first technique used while collecting data about how the current system work from paper-based approach. Farmers and other stakeholders have mentioned that the current system is helping but it still needs some improvement.

This technique helped to obtain accurate information since there was direct interaction with the respondents and they provided with relevant information.

The collected data was inspected, processed, and modeled with the goal of highlighting useful information, suggesting conclusions, and supporting decisions. In making the system requirements specification documents that was used as in input in designing the system.

Documentation

This technique involves systematic data collection from existing sources such as: books, statistics report, websites and so on. Reading and researching on various project documents from several sources helped to achieve the goal. The ideas and concepts gained have influenced and helped to shape the development of the project.

Research questions

During the study, the following research questions will guide data collection and interviews with stakeholders such as farmers and agronomists:

- How does the current system support farmers in managing and following up their agricultural activities?
- How efficient is the current system in terms of time consumption during data collection and processing?
- How will the introduction of a new or improved system positively impact both agronomists and farmers?

Expected Results

After the successful development, implementation, and testing of the system, a range of functional and operational outcomes are expected. These outcomes reflect both the direct impact on users and the broader contribution of the system to smart agriculture practices.

The system is expected to significantly improve the productivity of smallholder farmers by enabling them to make timely and informed farming decisions based on accurate and real-time data. By providing access to updated information on soil conditions, weather patterns, and crop status, farmers will be better equipped to plan agricultural activities such as planting, irrigation, fertilization, and harvesting, resulting in improved yields and reduced crop losses.

Another key expected outcome is enhanced decision-making through real-time farm data. The system will integrate data from sensors and user inputs to provide actionable insights that help farmers and agricultural officers respond quickly to environmental and crop-related changes. This real-time visibility reduces guesswork and supports precision agriculture approaches.

The system also aims to increase the adoption of affordable smart farming technologies among smallholder farmers. By providing a user-friendly and cost-effective solution, it encourages farmers to embrace digital agriculture tools, bridging the gap between traditional farming practices and modern agricultural innovations.

Furthermore, the system is expected to enhance communication between farmers and agricultural officers. This improved interaction will allow faster response to field issues such as pest infestations, disease outbreaks, unexpected weather changes, and soil degradation. Agricultural officers will be able to provide timely guidance and support, improving overall farm management and crisis response.

Overall, the system is expected to modernize agricultural information management processes by integrating data-driven decision-making, improving efficiency, and supporting sustainable farming practices. This transformation will ultimately contribute to improved food security, increased farmer resilience, and a more technology-driven agricultural sector.

Organization of the Report

This project report is divided into five main chapters as follows:

Chapter 1 General Introduction - gives an overall idea about the study, which includes Introduction, Background, Problem Statement, Motivation, Objectives, Scope, Methodology, and Expected Results.

Chapter 2 Analysis of the Existing System - gives details about the existing system, including problems faced in the current system and solutions to those problems.

Chapter 3 Requirement Analysis & Design of the Proposed New System - covers the design of the new system with its possible solutions through the UML methodology.

Chapter 4 Technical Realization of the Application - talks about technical aspects of how the application is developed, deployed, tested. Recommendations - gives recommendations after summarizing the whole study.

Chapter 5 Conclusion and Recommendations.

CHAPTER 2

ANALYSIS OF EXISTING SYSTEM

Introduction

In order to solve a problem effectively, a deep knowledge of the problem environment is needed. The analysis of a system should be done in the first place; that means looking into the existing system, detecting the problem areas and basing on the gathered information defining requirements for the new system. Otherwise, the result will definitely be bad.

Therefore, the analysis of a system is the investigation of the system, discovering problems and recommending solutions based on the obtained information. This chapter focuses on the methods of collecting data, Aguka Smart Farming, the structure and objectives of the system, as well as on gathering all information concerning the existing system.

Discovering the problems of a system and structuring information about it is the definition of system analysis. Another definition of system analysis would be the systematic breakdown of the system to its components. It is important to know how some things fit other elements, why certain components work or fail, and what works separately.

This procedure helps to learn the details of a system. System analysis can be performed while developing a system or analyzing an existing one. It is the same thing both ways.

In addition, system analysis plays an important role in identifying gaps between the current practices and the desired system performance. It helps researchers and developers understand how information flows within the existing system, how decisions are made, and where inefficiencies occur. Through this process, it becomes easier to propose appropriate improvements that align with user needs and technological advancements. As a result, system analysis forms a strong foundation for designing a more efficient, reliable, and user-friendly solution.

Furthermore, it ensures that system requirements are clearly defined before development begins. It also helps to reduce risks of failure during implementation. Ultimately, it improves the overall quality and success of the proposed system.

Description of Current System Environment

Historical Background

IMBARAGA is a National Farmer Organization established in 1992 which seeks to enhance socio-economic conditions of farmers estimated to be about 27,300 located in 25 of the 30 districts in the country. IMBARAGA currently operates as a Local NGO since initially it was set up as a Trade Union Organization. However, after the change of law that regulates operation of Local NGO which now no longer recognizes trade unions, IMBARAGA changed its status to become a local NGO with the same mandate, mission and vision of safeguarding interests of farmers. IMBARAGA was set up by farmers due to the challenges farmers had to face as a result of being voiceless. Due to the challenges, farmers came together with common goals of addressing those problems. IMBARAGA was formed out of the need of fighting for the interests of farmers.

In various periods of time, the federation of IMBARAGA has been partnering with various organizations including civil societies both locally and internationally. Such a partnership can be seen in terms of defending socio-economic rights of our members which include our role in the formulation of the Land Law of Rwanda, ensuring reasonable prices of potatoes among the farmers growing them in North and Northwest provinces, adding value to washed potatoes in the Northern Province, receiving recognition of our organization as one of professionals in the agricultural sector, development of our policies which we use as references in our advocacy & lobbying and gender issues.

Vision

The vision is to ensure that his/her member is “A professional farmer and change maker.”

Mission

The mission of Imbaraga is “To enhance the farmers’ professionalism through capacity building for productivity and competitiveness and protection of farmers’ social and economic rights through lobby and advocacy.”

Description of Current System

The farmers in the Imbaraga Farmers Organization currently use a system that relies on paper. They use notebooks and hard copy forms to write down information. The information they collect includes crops they produce, observations made during farm visits, and other notes related to farming activities.

This system helps them keep basic records and track what farmers are doing. However, it has several significant limitations. It takes a lot of time to manage and maintain, and data entry is done manually, which increases the workload for agronomists and extension officers. In addition, human errors are common when recording information by hand, which affects the accuracy and reliability of the data collected. It is also difficult to quickly locate specific information when needed.

The data is not stored in a centralized digital system, making it hard to analyze farmers' performance over time. As a result, generating reports and making informed decisions becomes slow and inefficient. This challenge is more critical when timely decisions are required to support farmers during planting seasons, pest outbreaks, or environmental changes.

Another major limitation is the lack of data sharing and communication efficiency. Since records are stored physically, they cannot be easily accessed by multiple stakeholders at the same time. This reduces collaboration between farmers, agronomists, and organization managers, leading to delays in response and coordination of agricultural activities.

Furthermore, the current system does not support real-time monitoring of farming activities. This makes it difficult to track progress, evaluate farm performance, or provide immediate feedback to farmers. Without digital tools, the organization cannot fully benefit from modern data-driven agriculture practices, which limits productivity and innovation.

Finally, the reliance on paper-based records also poses risks related to data loss and damage. Documents can be easily misplaced, destroyed, or deteriorate over time due to environmental conditions. This creates gaps in historical data, making long-term planning and analysis even more challenging for the organization.

Analysis of the Current System

The analysis conducted revealed critical weaknesses within the current system and the processes used to run the current operation. Various types of data were collected from farmers, agronomists, and existing records as a means to better understand how the system functions and to identify areas where improvements can be made. This involved reviewing daily practices, observing how information is recorded, and evaluating how data is stored and utilized for decision-making.

Farmers at the user level depend on a manual method for capturing and managing their daily farming activities. Farmers record their agricultural activities either individually or with assistance from field staff on paper forms, which are later submitted to field offices for processing. While this approach allows for basic data collection and minimal follow-up, it is generally slow, error-prone, and not suitable for long-term tracking or analysis of farming performance.

On the administrative side, the management of farmer records, generation of reports, and monitoring of agricultural activities becomes increasingly difficult due to the absence of a centralized digital system. The lack of integration between different records leads to duplication of work, delays in reporting, and reduced efficiency in service delivery. These challenges highlight the importance of adopting a more structured and automated approach to data management.

In addition, communication between farmers and agronomists is often delayed because information must physically move through paper-based channels. This slows down the response time when farmers need urgent support, such as pest control advice or weather-related guidance. As a result, decision-making is not always timely, which can negatively affect crop productivity and overall farm outcomes.

Finally, the reliance on manual processes reduces scalability as the number of farmers and data volume increases. As the organization grows, the workload on agronomists also increases significantly, making it harder to maintain consistent data quality and service delivery. This clearly indicates the need for a more efficient, digital, and integrated system to support modern agricultural management.

Modeling of The Current System

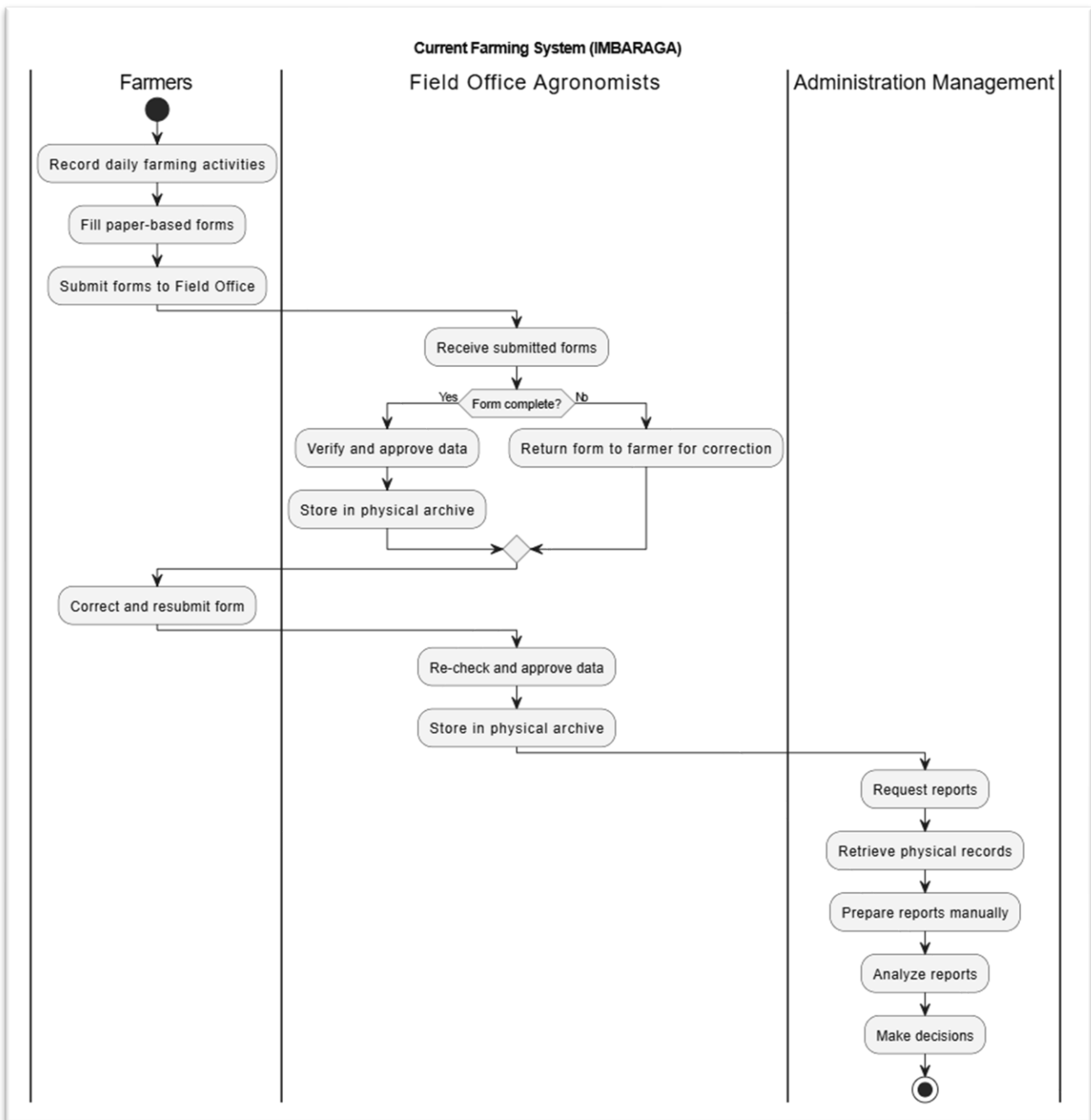


Figure 1: The current system modeling

Problems of the Current System

- **Performance**

Throughput

- Data volume is low as agronomists rely on manual input in paper logbooks. Slow processing speeds since agronomists have to physically check handwritten data.
- Massive backlog due lack of a centralized digital storage.
- Cannot provide real-time updates across multiple districts at the same time.

Response Time:

- Manual communication of weather alerts and seed planting instructions has created extended delays for farmers in receiving important information from their agronomists.
- Farmers will often have to wait for days or weeks until their agronomist can review records by hand before receiving feedback.
- The inability to have access to data in real-time delays emergency responses to pest outbreaks and/or crop disease.
- It is time-consuming for agricultural organizations to manually process and compile all farmer data for reporting purposes.

- **Information**

Input

- Manual writing is the primary method for recording data in notebooks, resulting in frequent inaccuracies and missing/altered essential information.
- Due to poor handwriting or not filling out paper forms correctly, field offices frequently miss significant points of content Materials.
- With no real time friendly way to enter both soil and weather data, accurate information is not able to be gathered on field conditions.

Output:

- Reports take a long time for managers to receive because they were generated at a slow rate which makes them contain outdated information.
- Farmers are not receiving timely and actionable feedback, such as irrigation alerts since output is not automatic in computation/production of reports.
- It is impossible to produce visual trends or summary reports, which frustrates and prevents IMBARAGA from having sufficient information to make data-based decisions.

• Storage

- Both paper and notebooks take up a lot of space, are prone to damage (fire, water), and can get lost.
- If paper records are lost or destroyed, it is almost impossible to recover historical data because digital backups are not available.

• Economics

- **Operational Costs:** There are high costs associated with buying paper, printing forms, and physically moving the documents for data collection.
- **Missed Opportunities:** Farmers lose money due to delayed bug control and soil issues that could have been prevented if they had timely data.

• Control

- **Monitoring and Supervision:** IMBARAGA headquarters has a hard time verifying that field data was recorded correctly and on time.
- **Accountability:** It's also hard to track who documented the data, leading to discrepancies in both farmer's information and resource allocation.

• Efficiency

- **Duplicate Process Redundancy:** Records are written in writer's notebooks before being copied to the authorized form, wasting time.
- **Operational Delays:** Administrative tasks come to a standstill because staff members have to wait for physical documents to travel from various districts.

Proposed Solutions

After Identifying and analyzing problem stated, here are the proposed solutions:

- **IoT-Integrated Smart Farming System:** An inexpensive hardware sensor system to monitor soil moisture, temperature, and nutrients, replacing guessing with exact data.
- **Multi-Channel Accessibility (Inclusive Design):** Farmers can utilize a dual access platform, available through a smartphone app for advanced users or a USSD interface for smallholder farmers employing basic phones. This ensures that we do not exclude any farmer depending on their access to technology.
- **Real-Time Data Collection and Automated Alerts:** Collecting data on the farm in real time will enable us to provide instant alerts to farmers regarding their crops' irrigation requirements or significant weather fluctuations.
- **Data Visualization and Analytics Dashboard:** With the gathered data, we are offering IMBARAGA management and farmers a digital visualization of soil trends and crop performance through an operational dashboard, enabling them to shift from traditional paper record-keeping to utilizing data for informed decision-making.
- **Automated Reporting System:** Utilizing a central module to automatically produce farm performance and administrative reports will streamline and enhance the consolidation of farm records from various districts once the records are entered into the individual notebooks.
- **Localized Educational Resources:** We are equipping farmers with tailored education in the app by offering them access to crop-specific advice and optimal practices in the Kinyarwanda language to enhance production methods and output.

In addition, the proposed system will enhance coordination between farmers and agronomists by ensuring faster communication and timely response to field issues. It will also reduce dependency on manual record-keeping, improve data accuracy, and support long-term agricultural planning through reliable digital records

System Requirements

Functional Requirements

1. SUPER ADMIN

REQ 1: The Super Admin shall have ability to login and logout from the system.

REQ 2: The Super Admin shall have ability to create, modify, delete and manage all the accounts (Admin, Cooperative, Extension Officer and Farmers).

REQ 3: The Super Admin shall have ability to assign roles and permissions to the users.

REQ 4: The Super Admin shall have ability to setup system wide configurations (Thresholds, Integrations, Notification rules).

REQ 5: The Super Admin shall have ability to access system audit logs and system activities monitoring.

REQ 6: The Super Admin shall have ability to backup and recover the system.

2. ADMIN

REQ 7: The Admin shall have ability to login and logout from the system.

REQ 8: The Admin shall have ability to manage the users within its assigned scope.

REQ 9: The Admin shall have ability to monitor the system usage and activities.

REQ 10: The Admin shall have ability to manage and validate the farm data.

REQ 11: The Admin shall have ability to generate system wide reports and financial reports.

REQ 12: The Admin shall have ability to configure the reporting and notification standards.

3. COOPERATIVE MANAGER

REQ 13: The Cooperative Manager shall have ability to login and logout from the system.

REQ 14: The Cooperative Manager shall have ability to manage its farmers.

REQ 15: The Cooperative Manager shall have ability to monitor aggregate farm data

REQ 16: The Cooperative Manager shall have ability to view the performance-based recommendation for farmers generated by system through AI for soil, weather and activity data.

REQ 17: The Cooperative Manager shall have ability to manage the resource allocation for its

members (Fertilizers, Equipments).

REQ 18: The Cooperative Manager shall have ability to generate Cooperative level reports.

REQ 19: The Cooperative Manager shall be able to manage group activities (meetings, events).

4. EXTENSION OFFICER

REQ 20: The Extension Officer shall have ability to login and logout from the system.

REQ 21: The Extension Officer shall be able to monitor assigned farms and farmers.

REQ 22: The Extension Officer shall be able to analyze soil, weather, and crop data.

REQ 23: The Extension Officer shall be able to provide recommendations and advisory messages to farmers.

REQ 24: The Extension Officer shall be able to track risks such as pests, diseases, and climate threats.

REQ 25: The Extension Officer have ability to generate advisory and monitoring reports.

5. FARMER

REQ 26: The Farmer shall have ability to login and logout from the system.

REQ 27: The Farmer shall have ability to set up and manage farm profile (location, crops, size).

REQ 28: The Farmer shall have ability to view soil data (moisture, temperature, nutrients).

REQ 29: The Farmer shall have ability to view weather conditions and forecasts.

REQ 30: The Farmer shall have ability to manage irrigation activities and view recommendations.

REQ 31: The Farmer shall have ability to record farm activities and track inputs.

REQ 32: The Farmer shall have ability to receive push notifications (alerts, recommendations).

REQ 33: The Farmer shall have ability to access crop and livestock management guidance.

REQ 34: The Farmer shall have ability to participate in community discussions and knowledge sharing.

Non-Functional Requirements

- **Performance**

- **REQ 35:** The system shall maintain an uptime of **99.9% annually**.
- **REQ 36:** The system shall recover from unexpected failures or crashes within **10 minutes** to maintain reliability.
- **Security**
 - **REQ 37:** The system shall implement **role-based access control (RBAC)** to restrict data access.
 - **REQ 38:** The system shall comply with Rwanda's data protection laws.
- **Usability**
 - **REQ 39:** The system shall provide a clean, intuitive interface that accommodates users with varying technical skills.
 - **REQ 40:** The system shall support multi-language functionality (Kinyarwanda, English, French) to serve diverse community stakeholders.
- **Compatibility**
 - **REQ 41:** The system shall be compatible with the latest versions of major web and mobile platforms (Android, iOS, Chrome, Firefox, Edge, Safari).
- **Security**
 - **REQ 42:** All data inside the system shall be protected against unauthorized access.

CHAPTER 3

REQUIREMENTS ANALYSIS AND DESIGN OF THE NEW SYSTEM

Introduction

This chapter presents the analysis and design of the proposed Aguka Smart Farming Kit Platform. The purpose of this phase is to identify system requirements and transform them into a structured design that can guide the implementation process. A thorough analysis at this stage is essential to ensure that the developed solution effectively addresses the challenges identified in the existing farming environment and aligns with user needs.

The chapter describes both functional and non-functional requirements of the system and explains how these requirements are represented using Unified Modeling Language (UML) diagrams. These diagrams provide a clear understanding of system behavior, user interactions, workflows, and data structures in a structured and visual manner. They also help in bridging the gap between system analysis and actual system development.

In addition, the chapter introduces the system architecture, database design, and other key design components that form the foundation of the proposed smart farming solution. These elements define how different parts of the system will interact, how data will be stored and processed, and how the system will operate as a whole.

Furthermore, this phase ensures that all technical and user requirements are clearly defined before implementation begins. It also helps in reducing design errors, improving system efficiency, and ensuring that the final solution is scalable, maintainable, and suitable for real-world deployment in the agricultural sector.

The design process also takes into consideration usability and accessibility to ensure that the system can be effectively used by farmers with different levels of technical knowledge. Emphasis is placed on simplicity of interaction, especially for users in rural areas. The system is also designed to support both mobile and low-bandwidth environments to enhance accessibility. In addition, security measures are considered to protect user data and ensure system reliability. Finally, the design aims to support future expansion and integration with other agricultural technologies.

Unified Modeling Language (UML)

The Unified Modeling Language (UML) is a standardized modeling language used to represent the structure and behavior of software systems. It consists of a set of visual diagrams that are used by system analysts, designers, developers, and other stakeholders to analyze and communicate how system components interact. In other words, UML serves as a communication tool that ensures a clear understanding among all parties involved in system design and implementation.

In software engineering, UML diagrams play an important role during the analysis and design phases of system development. Instead of relying only on text-based descriptions of system functionalities, workflows, and data structures, UML provides graphical representations that make complex system processes easier to understand and interpret.

In this project, UML diagrams were used to model the proposed system before its actual implementation. These diagrams provide a visual representation of interactions between different stakeholders, system processes, and data structures within the Aguka Smart Farming Kit Platform. They help illustrate how the system operates and define the roles and operations performed by different users.

The use of UML diagrams improves documentation quality and provides developers with a clear reference for implementation, testing, and maintenance. It also helps in identifying potential design issues early in the development process, ensuring that the final system is well-structured, efficient, and aligned with user requirements.

Additionally, UML supports better system planning by making it easier to validate requirements with stakeholders before development begins. It enhances communication between technical and non-technical users by presenting system logic in a simplified visual form. This contributes to reducing misunderstandings during development and improves the overall quality of the final solution.

Furthermore, UML diagrams help in identifying system dependencies and relationships between different components at an early stage. They also assist in improving system scalability by clearly defining how modules interact. In addition, UML provides a strong foundation for future system updates and maintenance by offering clear documentation of system design.

UML models

A variety of UML diagrams were applied to present the different elements of the Aguka Smart Farming Kit. Each diagram highlights a different aspect of the proposed solution and contributes to a deeper comprehension of its architecture.

First of all, the use case diagram was designed to reveal the core features of the proposed solution and represent interactions between the platform and its users. Thus, the use case diagram illustrates the main services provided by the system.

Secondly, the class diagram represents the static structure of the proposed solution and describes all the entities, their attributes, operations, and relationships. Class diagrams are especially helpful to create a logical design of the proposed application and its database.

Moreover, sequence diagrams were created to represent the communication between different components of the proposed system while performing some specific operations. This kind of diagram illustrates the order of message exchange.

Finally, activity diagrams are used to depict business processes in the proposed platform and perform workflow modeling. The diagram presents the sequence of operations, decisions, and actions.

In total, the mentioned UML models make up the basis for designing the Aguka Smart Farming Kit Platform.

Definitions and description of some UML concepts used

From a practical point of view, when applying the Object-Oriented technique, there will be certain concepts applied frequently. Let us see their definition, which is as follows:

A class: The OOP concept of a class is a program-code-template for creating objects. In addition to giving initial values to the properties of the class, a class specifies a set of methods (member functions) implementing class-specific behaviors.

Similar to an Abstract Data Type (ADT), classes contain both structure and behavioral data. However, unlike ADTs, classes can be defined in terms of other classes via inheritance. It can be said that a class serves as a template for creating objects.

There are three main components to a class, which are:

- The first part helps in creating the name of the class.
- Secondly, the attributes of the class are written down in the middle.
- Later, there will be a list of the methods that the class might apply.

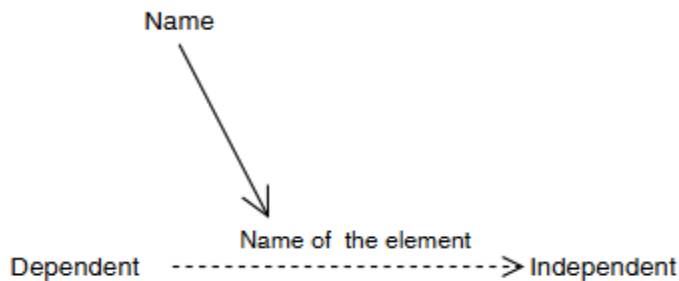
An attribute refers to the property of a class that helps in defining a range of values that instances of the property might have.

A method is a piece of code that will execute when called to perform tasks. Parameters may be passed to a method. The primary purpose of methods is to perform certain actions.

A table is a kind of table that stores information in rows and columns. It may be used for storing as well as displaying information. For instance, databases have information stored in the form of tables in order to be able to access the information from rows in an easy way. Tables are used in websites for displaying multiple rows of information on a website page. Spreadsheets, on the other hand, combine both aspects of a table. A table consists of rows and columns. The table specifies a particular number of columns, but not rows.

Relationships: A relationship in UML is a link between two or more UML model elements that gives additional information to the model. It may either take the form of a diamond shape or, much simpler, a line joining the two elements. Verbs are used to name the relationships. A relationship helps us to demonstrate how two or more things are connected on the model. Relationships enable us to capture semantically meaningful connections between things on the model. In OOM, we mainly use three types of relationships: dependencies, generalizations, and associations.

Dependency: A dependency is a relationship that shows that changes in one model element will result in changes in another model element. It shows that one model element uses the information and services of another model element without necessarily the vice versa. Dependency is depicted as a dashed-line arrow as demonstrated below. The arrow tail represents the independent element while the arrow head represents the dependent model element

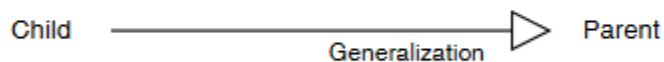


Dependency relationship is used to represent the dependency relationship between two elements of a system.

Generalization Relationship: A generalization relationship defines that a child model element derives from a parent model element. Although a parent model element may have one or many child elements, the child elements might also have one or many parents, except the fact that a parent will always have more than one child element. Object of a child element can be assigned to a variable or parameter of type parent but vice versa assignment is not allowed. Child will inherit all the attributes and operations owned by the parent(s). In UML 2.0 several classes form a generalization set of another class.

Generalization relationships occur in class, component, and use-case diagrams.

A generalization relationship is depicted using an arrow with hollow arrow head as shown in the figure below:

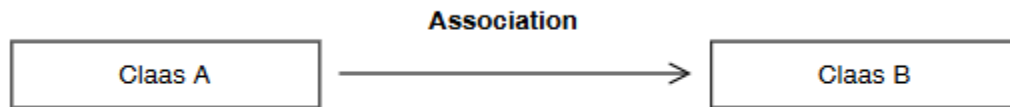


Generalization is used to describe parent-child relationship of two elements of a system.

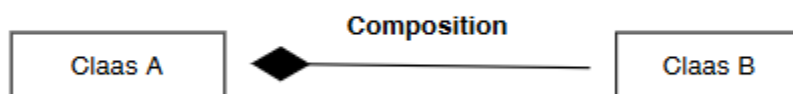
Association: An association relationship is a structural relationship between two model elements depicting the connection and navigability between objects of one classifier (actor, use case, class, interface, node, or component) and objects of another classifier. If there is an association relationship between two classes then objects of one class relate with objects of other class. An

association relationship may be specialized into an aggregation relationship or even further strengthened to become a composition relationship.

An association relationship is defined as follows:

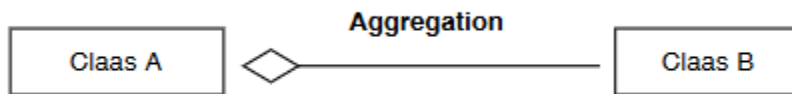


Composition: a composition relationship depicts whole-part relationship and is a form of aggregation. A composition relationship specifies that life span of the part classifier depends upon life span of whole classifier. Composition relationship is the strong association relationship. If an association relationship depicts ownership of object by other object and object exists only if owned by other object then such an association is called composition relationship. For example, let us consider that there is human body having heart. In this case human object owns the heart and heart does not exist without human object. Composition is a restricted form of aggregation where two objects are highly dependent on each other. It is a 'part of' relationship. A composition relationship involves two objects which are dependent on each other. When composition occurs between two objects then object of composed cannot exist without another object.



Aggregation: Aggregation with respect to objects is defined as “HAS-A” relationship between two objects that are related to each other. For instance, each sector comprises several sites. Aggregation is referred to the presence of the set of child objects inside the parent class. Aggregation is a weak association. Association is termed aggregation when both objects can live on their own. For example, an object of “team” and another object of “player”. The former contains several players while the latter can live without the team.

An aggregation is defined as follows.



Process of software development

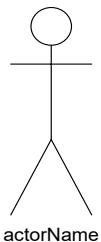
The process of software development can be described as the process where tasks are divided among various phases or stages in order to organize their completion more effectively in software engineering. This is regarded as part of the systems development life cycle. The process can also entail the establishment of certain deliverables that should be delivered as results from a particular phase of software development. The objective in my situation would be to develop software that fulfills the user's need.

In the field of developing applications, there exist several techniques that can be applied in achieving the goal of software development. In technical terms, there can be the implementation of such processes as waterfall development, parallel development, phased development, prototyping and throwaway prototyping and Agile.

Use case diagram components description

Actor

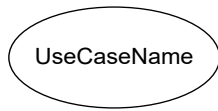
While engaging directly with the system being developed, the external entity adopts the role of an actor. The role can involve the duty performed by the user or a system interacting with the current system.



Use case

A use case is a scenario that includes events and/or actions required to define interaction between an actor ("role" in UML) and a system for accomplishing a particular goal. An actor can be a

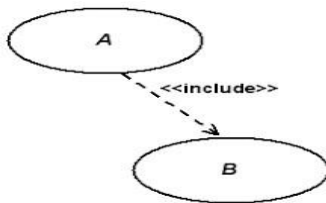
person, an outside system, or a point in time. Besides, use case is just a description of a set of interactions between a user and a system. Below is an illustration of this concept.



Relationship

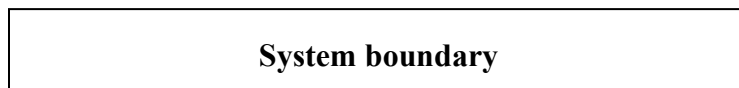
Meaningful relationship between actors and use cases (which means the UML association).

Include relationship: The inclusion of the functionality of one-use case with another



System boundary

A box used to indicate the boundary of the system under consideration within the use case.



The following figure illustrates the operations of the new system with stakeholders via the use case diagram.

Design of the new system

System design is the process through which the functionalities identified in the analysis phase will be designed. The aim of the design process is to develop a technical solution that meets the functional requirements of the system. The system design phase is where the architecture, components, modules, interfaces, and data of a system are defined to meet the requirements.

Use Case Diagram for Administrative Users

The below use case diagram shows the interactions that occur between the administrative users and the Aguka Smart Farming Kit. The tasks of Super Administrators and Administrators include managing users, setting up the system settings, validating farms, monitoring the actions taking place on the platform, producing reports, ensuring security on the platform, and facilitating the entire process.

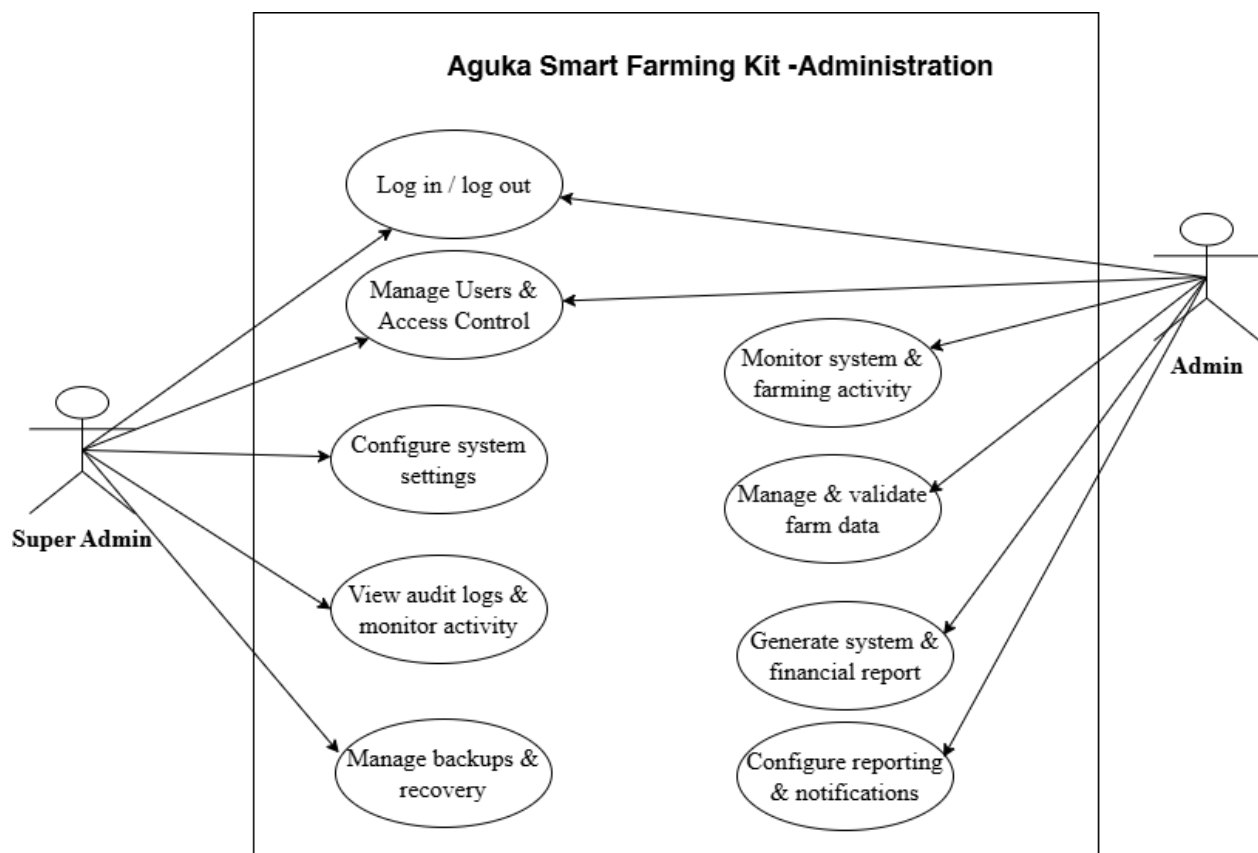


Figure 2: administration use case diagram

The administration use case diagram illustrates the responsibilities of Super Administrators and Administrators within the Aguka Smart Farming Kit. These actors are responsible for maintaining system integrity, managing users, validating farm information, monitoring platform activities, generating reports, and enforcing security policies. The diagram demonstrates how administrative functions are centralized to ensure effective governance and smooth operation of the platform.

Operational Users Use Case Diagram

The below use case diagram shows how operational users interact with the Aguka Smart Farming Kit. Farmers use the platform to manage farms, monitor environmental conditions, and receive recommendations. Extension Officers provide technical support and monitor farm performance, while Cooperative Managers oversee member farmers, coordinate resources, and generate cooperative reports.

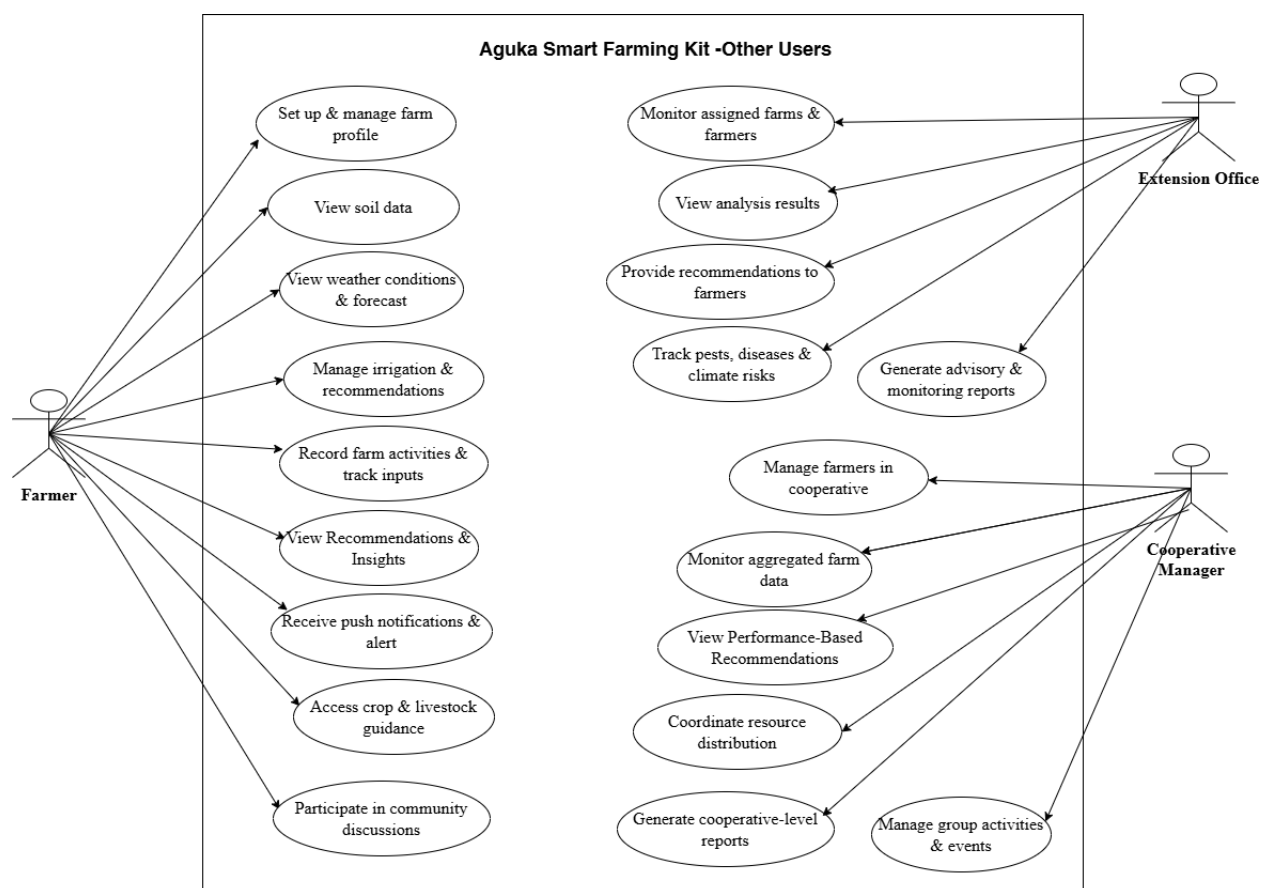


Figure 3: operational users use case diagram

The operational users use case diagram represents the interactions between farmers, extension officers, cooperative managers, and the system. These actors perform day-to-day agricultural activities including farm management, environmental monitoring, activity recording, recommendation management, reporting, and advisory services. The diagram highlights how the platform supports collaborative decision-making and smart farming practices through role-based functionalities.

Use Case description

The use case description explains how a use case functions and describes all requirements that should be met when performing the use case. It includes:

- **Name:** Name of a use case.
- **Description:** Goal of a system.
- **Actor:** Participant in the use case.
- **Pre-condition:** System condition before initiating the use case.
- **Post-condition:** Conditions after completing the use case.
- **Normal flow:** Real flow of a use case.
- **Alternative flow:** Possible actions in case a normal flow fails.

Login use case Description

Use Case Number:	UC-01
Use Case Name:	Login/ Logout
Actor:	Super Admin, Admin, Farmer, Extension Officer, Cooperative Manager
Description:	Allows users to login into the system with role-based access control.
Pre-condition:	Users should have a registered account with valid credentials.

Post-condition:	The system generates a message stating that the login was successful.
Normal Flow:	1. User accesses the system and initiates the login process. 2. The system shows the login form. 3. User fills in the login form with the right data and presses the submit button. 4. System checks the credentials entered by a user. 5. In case of credentials' correctness, the system logs in a user. 6. The system shows a message that says login was successful.
Alternative Flow:	A. . In case login credentials were incorrect, the system will display an error message. B. The user corrects the errors and re-sends the login form.

Table 1: use case for login

Manage Farm Profile

Use Case Number:	UC-02
Use Case Name:	Manage Farm Profile
Actor:	Farmer
Description:	Allows a farmer to set up, edit, and manage their farm profile including location, size, crop types, and irrigation methods.
Pre-condition:	Farmer must be logged in and have no existing farm profile (or be editing an existing one).
Post-condition:	Farm profile is saved, and soil/weather data can now be linked to the farm.

Normal Flow:	1. Farmer selects "Farm Profile" from the dashboard. 2. System displays a form with fields (farm name, size, location, crop type). 3. Farmer enters/edits details and submits. 4. System validates the inputs. 5. System saves the farm profile. 6. System confirms success.
Alternative Flow:	A. if required fields are missing, system prompts user to complete them. B. User corrects errors and resubmits.

Table 2: use case for manage farm profile

View Soil & Weather Data

Use Case Number:	UC-03
Use Case Name:	View Soil & Weather Data
Actor:	Farmer, Extension Officer, Cooperative Manager
Description:	Allows users to view real-time and historical soil data (moisture, pH, nutrients) and weather conditions (temperature, rain, forecast).
Pre-condition:	Users must be logged in with an active account.
Post-condition:	User views graphs and metrics of soil/weather data for their farm or assigned farms.
Normal Flow:	1. User selects "Soil Data" or "Weather" from dashboard. 2. System retrieves latest sensor data. 3. System displays dashboard with graphs, gauges, and 7-day forecast. 4. User can filter by date range or sensor type.

	5. System updates display accordingly.
Alternative Flow:	A. If sensors are offline, system shows "Last known data" with timestamp. B. If no data exists, system shows "No data available".

Table 3: use case for view soil & weather data

Manage Users & Access Control

Use Case Number:	UC-04
Use Case Name:	Manage Users & Access Control
Actor:	Super Admin, Admin
Description:	Allows administrators to assign system roles (Farmer, Extension Officer, Cooperative Manager) and set specific permissions (view, edit, approve).
Pre-condition:	Admins must be logged in. Target user account must exist.
Post-condition:	User account is updated with new role and permissions.
Normal Flow:	1. Admin accesses "User Management". 2. System displays list of users. 3. Admin selects a user and clicks "Assign Role". 4. System displays role options and permission checkboxes. 5. Admin selects role and permissions, then submits. 6. System validates the assignment (e.g., Super Admin cannot be demoted). 7. System saves and confirms.
Alternative Flow:	A. If user already has a role, system prompts for confirmation to override. B. If permissions conflict, system shows warning.

Table 4: use case for manage users & access control

Manage Irrigation Recommendations

Use Case Number:	UC-05
Use Case Name:	Manage Irrigation Recommendations
Actor:	Farmer, Extension Officer
Description:	Allows users to view AI-driven irrigation recommendations based on soil moisture, weather forecast, and crop type. Farmer can accept/apply or override.
Pre-condition:	Farm profile must exist. Soil moisture and weather data must be available.
Post-condition:	Recommendation is recorded, and if accepted, an irrigation schedule is created.
Normal Flow:	1. User selects "Irrigation Recommendations". 2. System analyzes current soil moisture and 3-day forecast. 3. System displays recommendation (e.g., "Irrigate 20 min tomorrow at 6 AM"). 4. User can Accept, Reject, or Modify. 5. If accepted, system logs the action and adds to farm calendar.
Alternative Flow:	A. If soil moisture is sufficient, system shows "No irrigation needed". B. If user modifies recommendation, system saves custom schedule.

Table 5: use case for manage irrigation recommendations

Record Farm Activities & Track Inputs

Use Case Number:	UC-06
Use Case Name:	Record Farm Activities & Track Inputs
Actor:	Farmer
Description:	Allows farmers to log daily activities (planting, fertilizing, harvesting) and track input usage (seeds, fertilizer, water).
Pre-condition:	Farmer must be logged in. Farm profile must exist.
Post-condition:	Activity is saved and input inventory is updated.
Normal Flow:	1. Farmer selects "Farm Activities" → "Add New". 2. System displays activity form (type, date, quantity, notes, photo). 3. Farmer fills details and submits. 4. System validates input (e.g., quantity > 0). 5. System saves activity and deducts inputs from inventory.
Alternative Flow:	A. If input quantity exceeds inventory, system shows warning. B. If photo is too large, system compresses it.

Table 6: use case for record farm activities

Monitor Assigned Farms

Use Case Number:	UC-07
Use Case Name:	Monitor Assigned Farms
Actor:	Extension Officer
Description:	Allows extension officers to view a dashboard of all farms assigned to them, monitor soil/weather/crop health, and identify at-risk farms.

Pre-condition:	Extension officer must be logged in and have farms assigned by Admin.
Post-condition:	Officer views a summary list and can drill into individual farm details.
Normal Flow:	1. Officer selects "My Assigned Farms". 2. System displays a list of farms with key metrics (soil moisture, last activity, alerts). 3. Officer clicks on a farm to view full details. 4. System validates input (e.g., quantity > 0). 5. System displays farm-specific data and history.
Alternative Flow:	A. If no farms are assigned, system shows "No farms assigned". B. If a farm has active alert, it is highlighted in red.

Table 7: use case for monitor assigned farms

Generate Cooperative-Level Reports

Use Case Number:	UC-08
Use Case Name:	Generate Cooperative-Level Reports
Actor:	Cooperative Manager
Description:	Allows cooperative managers to generate aggregated reports on member farmer performance, yield estimates, resource distribution, and input usage.
Pre-condition:	Manager must be logged in and have farmers linked to the cooperative.
Post-condition:	A downloadable report (PDF/Excel) is generated and saved to the system.
Normal Flow:	1. Manager selects "Reports" → "Cooperative Report". 2. System displays filter options (date range, farmer, crop type). 3. Manager selects filters and clicks "Generate".

	4. System aggregates data from all member farms. 5. System displays a preview of charts and tables. 6. Manager can download as PDF or Excel.
Alternative Flow:	A. If no data exists for the selected period, system shows "No data available". B. If report generation takes >10 seconds, system shows progress bar.

Table 8: use case for generate cooperative-level report

View Audit Logs

Use Case Number:	UC-09
Use Case Name:	View Audit Logs
Actor:	Super Admin
Description:	Allows the Super Admin to view a complete audit trail of all user actions, logins, data changes, and system events for security and compliance.
Pre-condition:	Super Admin must be logged in.
Post-condition:	Audit logs are displayed with filters and export options.
Normal Flow:	1. Super Admin selects "System" → "Audit Logs". 2. System retrieves recent logs (user, action, timestamp, IP). 3. System displays logs in a table with search/filter options. 4. Super Admin applies filters (date, user, action type). 5. System updates the displayed logs.
Alternative Flow:	A. If logs exceed 1,000 records, system paginates. B. If export is requested, system generates a CSV file.

Table 9: use case for view audit logs

Receive Push Notifications & Alerts

Use Case Number:	UC-010
Use Case Name:	Receive Push Notifications & Alerts
Actor:	Farmer, Extension Officer, Cooperative Manager
Description:	Allows users to receive real-time alerts on weather warnings, pest/disease risks, irrigation needs, and system updates.
Pre-condition:	User must be logged in (or have token registered). Notification permissions must be enabled.
Post-condition:	User receives a push notification on mobile/web.
Normal Flow:	1. System triggers an alert (e.g., heavy rain forecast). 2. System identifies affected users based on farm location. 3. System sends a push notification to their device. 4. User taps notification → opens relevant screen in the app. 5. System displays farm-specific data and history.
Alternative Flow:	A. If user is offline, notification is queued for next login. B. If user dismisses, it remains in an "Inbox" section.

Table 10: use case for push notification

View Recommendations & Insights

Use Case Number:	UC-011
Use Case Name:	View Recommendations & Insights
Actor:	Farmer

Description:	When a farmer enters data into the system (soil readings, weather data, farm activities, crop information), the system automatically processes this input and displays AI-generated recommendations, insights, and predictions as outputs/results.
Pre-condition:	Farmer must be logged in. Farm profile must exist.
Post-condition:	Farmer views personalized AI recommendations generated from their input data.
Normal Flow:	1. Farmer enters data into system (soil reading, activity log, crop update) 2. System automatically processes input through AI Engine 3. System displays outputs/results on dashboard: • Irrigation recommendations based on soil moisture • Fertilizer advice based on soil nutrients • Pest alerts based on weather conditions • Yield predictions based on crop growth stage 4. Farmer can Accept, Dismiss, or Download recommendation 5. If accepted, system adds to farm calendar
Alternative Flow:	A. If input data is incomplete, system shows "Add more data for better recommendations" B. If urgent alert detected, system sends push notifications immediately regardless of app usage

Table 11: use case for view recommendations & insights

View Performance-Based Recommendations

Use Case Number:	UC-012
Use Case Name:	View Performance-Based Recommendations

Actor:	Cooperative Manager
Description:	The system automatically compares performance metrics across all member farmers and generates AI-powered recommendations for each farmer based on their performance gaps. The manager simply views the results - the system does all the analysis automatically.
Pre-condition:	Manager must be logged in. Multiple farmers linked to the cooperative must have sufficient historical data.
Post-condition:	Manager views comparison report with automatically generated AI recommendations for each farmer.
Normal Flow:	1. Manager selects "Performance Comparison" from dashboard 2. The system automatically retrieves all farmers under the cooperative, fetches performance metrics such as yield, efficiency, and crop health, calculates cooperative averages, identifies the top three performers and bottom three performers, and generates personalized recommendations for each farmer. 3. The system displays a comparison table showing each farmer's metrics compared to the cooperative average, performance trends indicated by arrows for improving or declining, and an AI Recommendations column with automatic suggestions. 4. Manager can click "Send" to push recommendations to farmers via notification 5. System tracks which recommendations were sent
Alternative Flow:	A. If insufficient data, system shows "Need at least 3 months of data for 3+ farmers" B. Manager can export comparison report (PDF/Excel) with AI recommendations included

Table 12: use case for view performance-based recommendations

View Analysis Results

Use Case Number:	UC-13
Use Case Name:	View Analysis Results
Actor:	Extension Officer
Description:	Allows an Extension Officer to view AI-generated analysis results for any assigned farm, combining soil readings, weather data, and crop information into actionable insights and risk flags.
Pre-condition:	Extension Officer must be logged in and have at least one farm assigned. Farm must have recent soil and weather sensor data.
Post-condition:	Officer views AI analysis panel showing soil health status, weather risk flags, crop condition assessment, and recommended actions.
Normal Flow:	1. Officer selects a farm from "My Assigned Farms" list. Officer clicks "View Analysis Results". 2. System retrieves latest soil, weather, and crop data. AI engine analyses data against optimal thresholds for the registered crop. 3. System displays colour-coded analysis: Green (normal), Amber (monitor), Red (critical). 4. System shows recommended actions (e.g., "Irrigate within 24 hours", "Pest risk detected"). 5. Officer can click "Send Advisory" to forward recommendations to the farmer.
Alternative Flow:	A. If insufficient sensor data exists, system shows "Not enough data for analysis — check sensor connection." B. If all readings are within normal range, system shows "Farm health is good — no action required."

Table 13: use case for view analysis results

Class Diagram

A class diagram is a structural diagram showing classes, their attributes, operations, and class associations to illustrate how the system is organized (Ijaz, 2020). A class defines the methods and attributes of an object, representing a specific element in a program or unit of code that describes its structure and behavior. In object-oriented programming, class diagrams are used to visualize relationships and dependencies between different classes within the Unified Modeling Language (UML). All object-oriented systems can utilize class diagrams to model system structure and interactions effectively. The classes are grouped based on similarities, shared responsibilities, and functional roles within the system architecture.

The class diagram of the Aguka Smart Farming Kit represents the object-oriented structure of the platform. The User class acts as a parent (superclass) for several specialized user types, namely Farmers, Extension Officers, Cooperative Managers, Administrators, and Super Administrators. These relationships demonstrate inheritance, where each subclass inherits common attributes and behaviours from the User class while also having its own specific functionalities tailored to its role in the system.

The diagram also illustrates associations between core system components such as farms, sensors, soil readings, weather data, recommendations, and reports. These relationships ensure smooth data flow and integration across the platform, enabling real-time monitoring and decision-making in agricultural operations. For example, a Farm can have multiple Sensors, each Sensor generates SoilReadings and WeatherData, which are then processed to produce Irrigation Recommendations and Reports.

Furthermore, the class diagram provides a static view of the Aguka Smart Farming Kit, clearly presenting system entities, their attributes, operations, and relationships. It serves as a foundation for both database design and software implementation by defining how different modules interact and exchange information within the system. This helps developers maintain consistency between the system design and its actual implementation.

Main entities in the system include User, FarmerProfile, Farm, Sensor, SoilReading, WeatherData, Crop, IrrigationRecommendation, Notification, Report, and ActivityLog. Each of these entities

plays a crucial role in ensuring that the system captures, processes, and delivers meaningful agricultural insights to users.

In addition, the class diagram supports system scalability and maintainability by allowing new modules or features to be integrated without disrupting existing structures. It also enhances understanding among developers and stakeholders by providing a clear visual representation of system architecture, making it easier to identify relationships, dependencies, and potential improvements in the Aguka Smart Farming Kit.

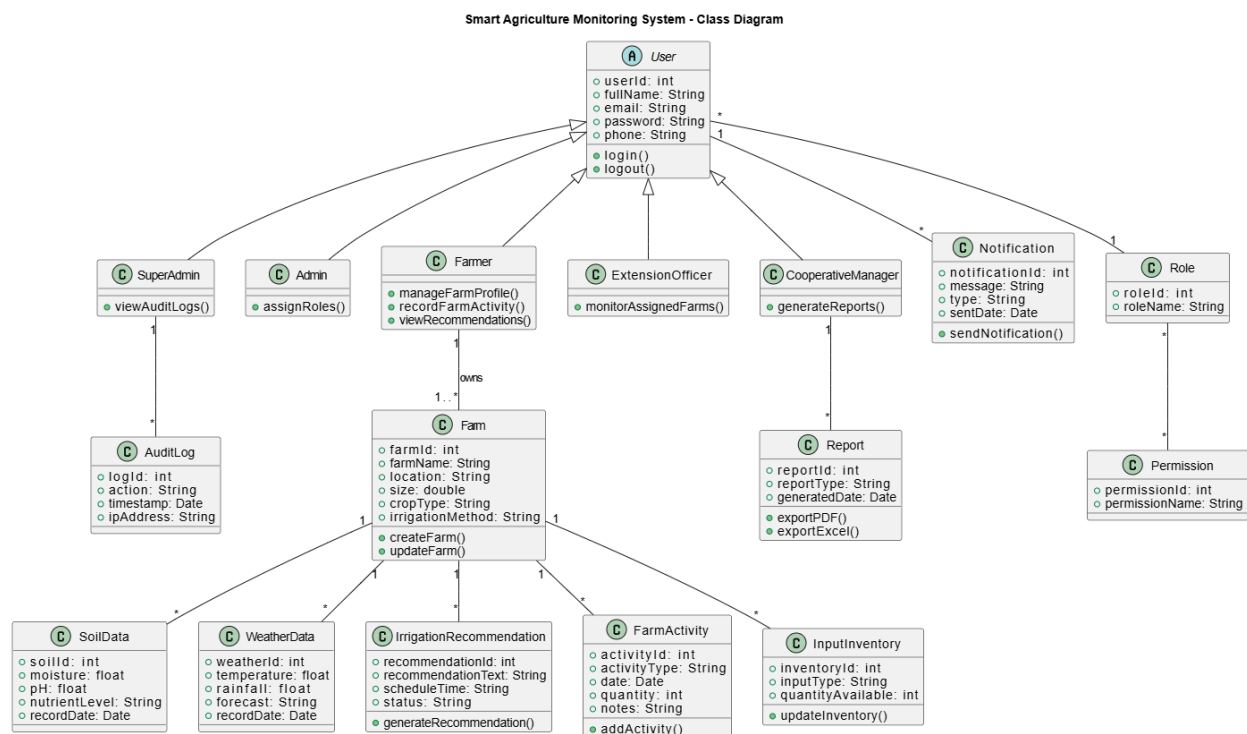
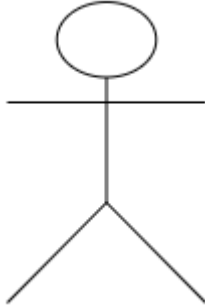
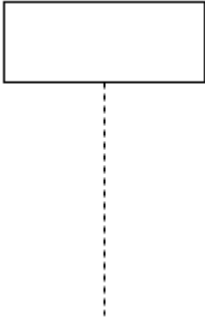
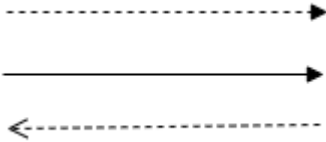


Figure 4: class diagram

Sequence Diagram

Practically, when we understand for example the sequence of numbers, we understand the order in which numbers are arranged. Therefore, in software engineering we have the order in which interactions between users, web servers, database and browser are followed one another. A sequence diagram is one of the types of diagrams that are widely used by developers. It shows the interaction of objects within a single use case. Sequence diagram represents the way of how different parts of the system cooperate with each other for executing a function, as well as the order in which the interactions take place while a specific use case takes place.

The Elements of a sequence diagram

Term / definition	Symbol
➤ Actor: ➤ Is an individual who uses software application. Is a person or a system that benefits from and is external to the system. ➤ Takes part in the sequence through sending or receiving messages. ➤ Located at the top of the sequence diagram.	
➤ Object Life Line: ➤ Takes part in the sequence through sending or receiving messages. ➤ Located at the top of the sequence diagram.	
➤ Activation: ➤ Is a long rectangle placed on top of a life line. ➤ It denotes when an object is sending or receiving messages.	
➤ Message: ➤ Communicates information from one object to another object.	

➤ Operation is marked as sent and a solid arrow, while return is marked with the value and dashed arrow.	
--	--

Table 14: Sequence diagram elements

Sequence Diagram for User Login

The User Login Sequence Diagram shows the interaction of the user, the login interface, authentication service and the database in the process of authenticating the user. The process starts when the user supplies login credentials, like email/username and password to the system interface. The login request is passed over to the authentication component, where it validates the supplied credentials by comparing them to the entries in the database. In case the credentials are correct, the system retrieves the user's role and access privileges, establishes a secured session and redirects the user to the right dashboard depending on his/her role: Farmer, Extension Officer, Cooperative Manager, Administrator or Super Administrator. In case the credentials are not valid, error message is sent to the user. It is necessary for ensuring secure access control, protecting the agricultural data and ensuring that users only access authorized functionalities.

Importance of the Sequence Diagram

This particular sequence diagram is an illustration of the process through which information is exchanged among system components while logging in. It explains how different elements of the authentication and authorization process work to confirm the identity of a user as well as provide role-based access control. This sequence diagram will assist the developer in understanding system interactions and implementing security requirements in Aguka Smart Farming Kit.

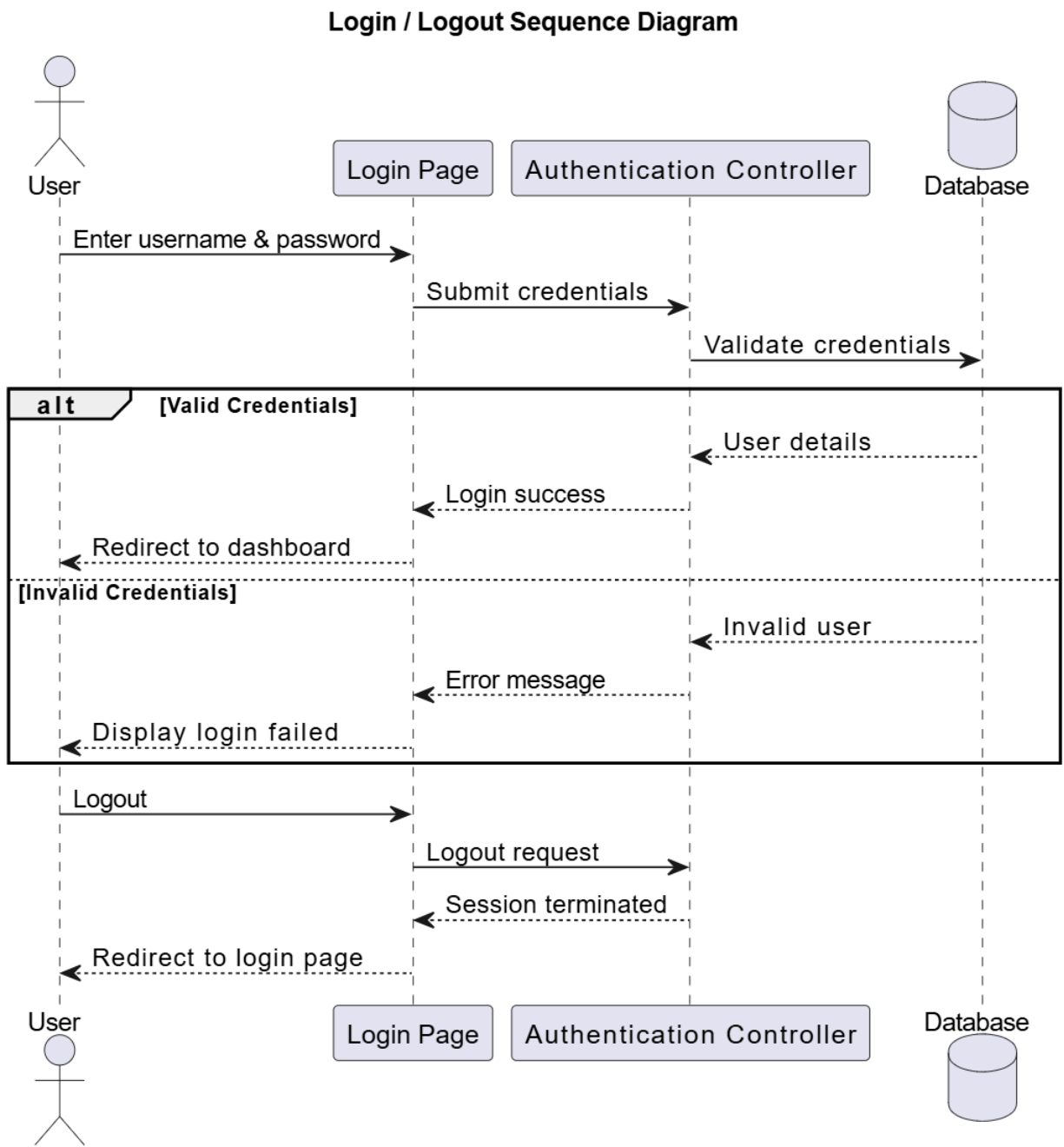


Figure 5: User Login Sequence Diagram

This sequence diagram demonstrates a secure and straightforward authentication mechanism where credential validation is handled server-side through the Authentication Controller, ensuring that user data is verified directly against the Database before any access is granted. The clear

separation between the Login Page, Authentication Controller, and Database reflects a layered architecture that promotes both security and maintainability within the Aguka platform.

Sequence Diagram for Manage Farm Profile

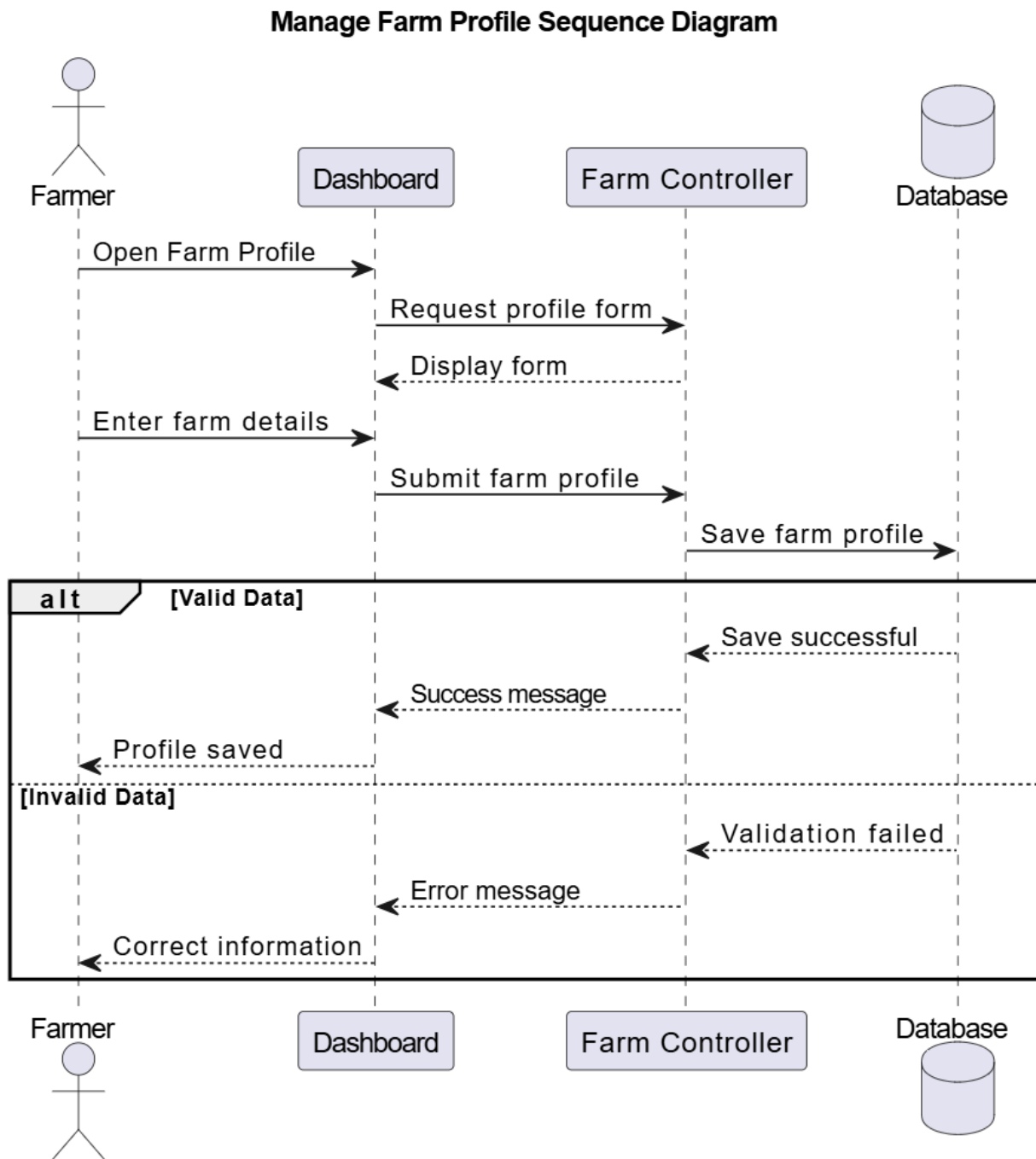


Figure 6: Manager Farm Profile Sequence Diagram

Above is the sequence diagram that highlights the effectiveness of the farm profile management process that features server-based validation process in order to ensure accuracy and completeness of the data within the Farm profiles that will be stored in the Database. From this sequence diagram one can observe an efficient division of roles between the Dashboard, the Farm Controller, and the Database where the latter acts as the central element in terms of data validation and database interactions with the use of layered architecture of the Aguka smart farming kit. Efficient validation of the data in the platform helps in generating reliable results of soil analysis and weather predictions for each farm.

Irrigation Recommendation Sequence Diagram

Below is the sequence diagram that illustrates the interactions between the farmer, the Aguka Smart Farming Kit, IoT sensors, recommendation engine, and the database during the irrigation recommendation process. The process starts when the farmer requests irrigation guidance through the system. In turn, the platform collects the real-time environmental data which includes soil moisture, soil temperature, humidity, and weather forecast data collected by means of IoT sensors and external weather services. After that, the recommendation engine analyses the data collected against the rules and intelligent decision-making algorithms in order to generate the appropriate irrigation schedule and water quantity/time.

Significance of the Sequence Diagram

This sequence diagram demonstrates the chronological flow of messages exchanged among system components during the irrigation recommendation process. It highlights how real-time sensor data, environmental conditions, and analytical services work together to generate accurate irrigation recommendations. The diagram also provides a clear understanding of system interactions, supports software implementation, and serves as a reference for testing and validating the irrigation recommendation functionality within the Aguka Smart Farming Kit.

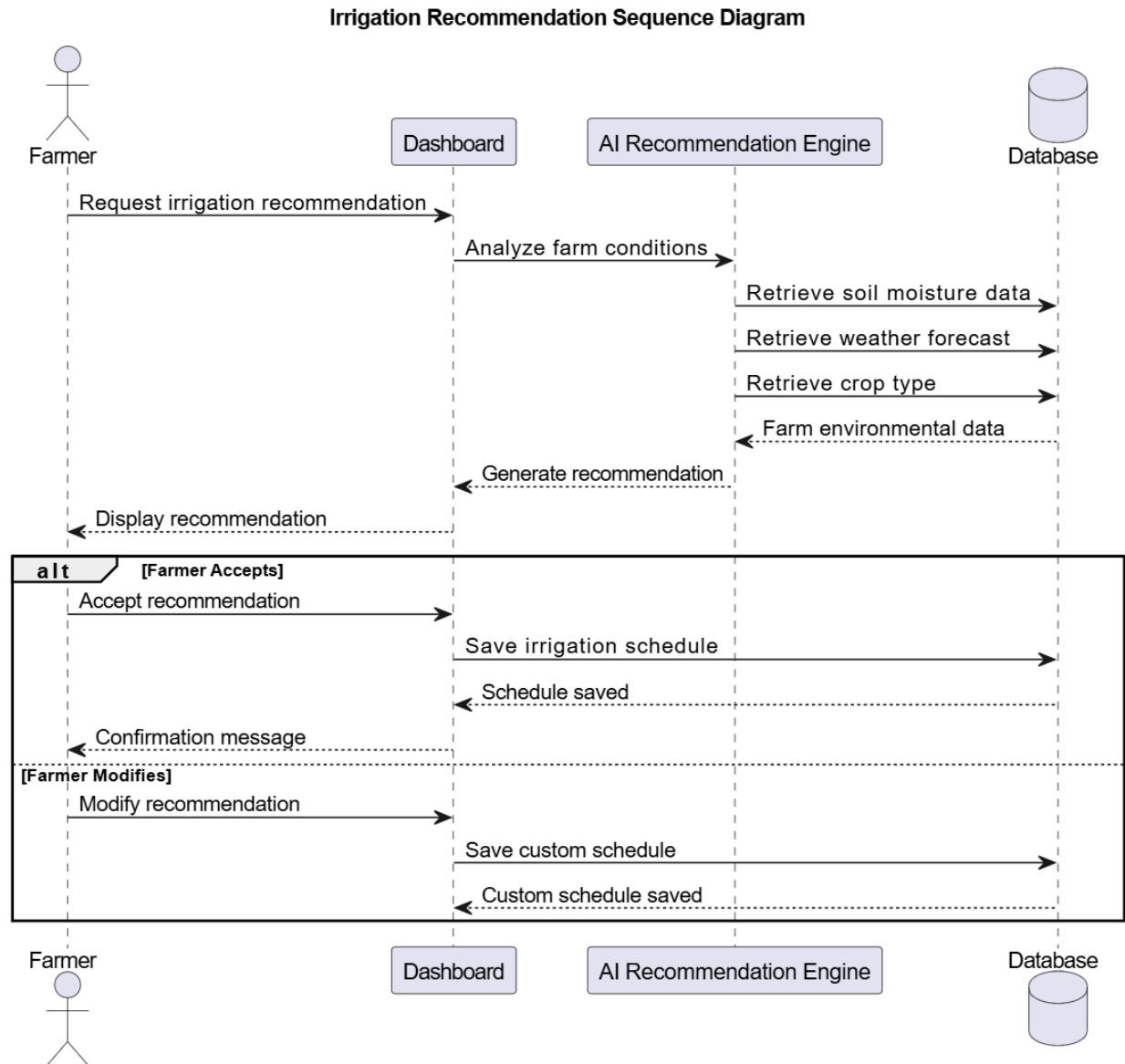


Figure 7: Irrigation Recommendation Sequence Diagram

The above sequence diagram reveals the intelligent nature of the Aguka Irrigation Module, which is built around the use of an innovative AI Recommendation Engine that integrates live soil moisture data, accurate hyperlocal weather prediction, and specific crop needs to provide farmers with optimal irrigation instructions. By integrating both paths - of acceptance and modification of the AI recommendation - the Aguka platform enables farmers to either completely rely on the AI recommendation or modify it based on their personal experience and field observations. Thus, although the platform provides recommendations based on data, the farmer himself is left with the

final decision on what to do, making the system less restrictive and fully in line with the principles of Data-driven Agriculture.

Sequence Diagram for View Soil & Weather Data

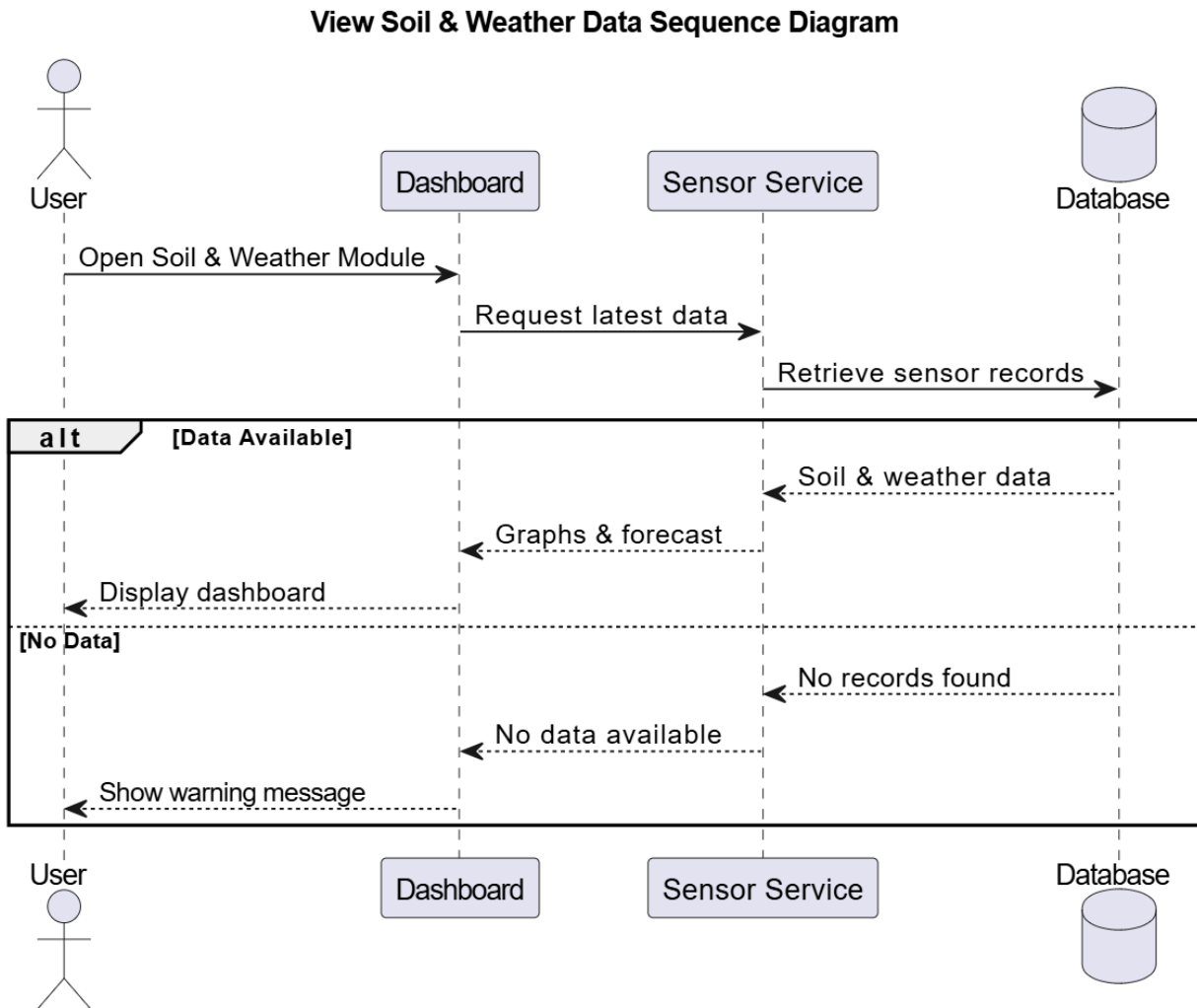


Figure 8: View Soil & Weather Data Sequence Diagram

The following sequence diagram explains how the Aguka platform implements real-time environmental monitoring through its efficient and dynamic process of retrieving the data. In the above diagram, Sensor Service represents the vital interface between the Database records and the Dashboard from which sensor data will be transformed into visual information that farmers and extension officers can understand. The presence of a no data handling option highlights the platform's ability to cope with scenarios where there is no data on the sensors, and yet users will

get meaningful information about their farm operations at all times. It should be noted that in a country such as Rwanda where farmers reside in remote rural areas, it is vital to keep an open line of communication with users.

Sequence Diagram for Cooperative Report

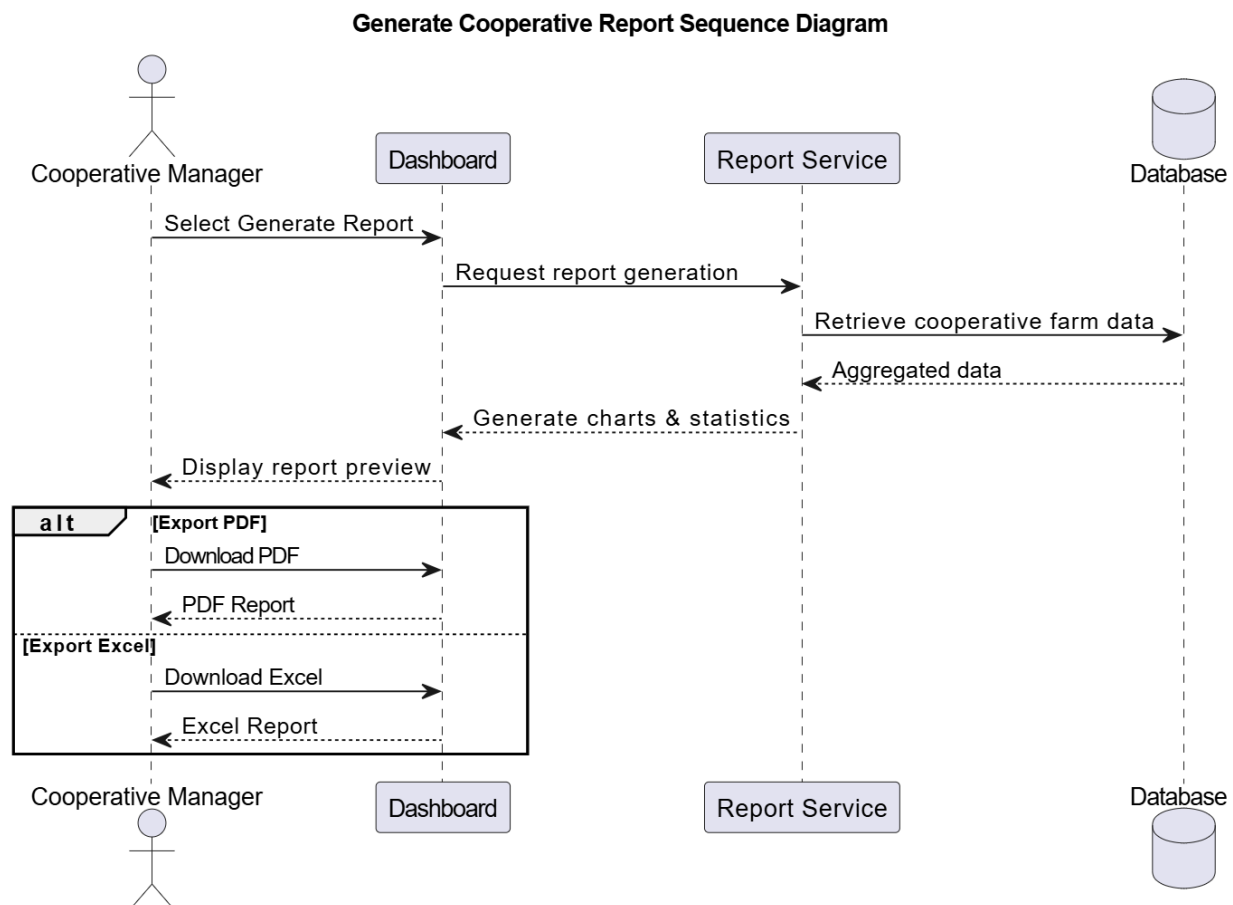


Figure 9: Generate Cooperative reports Diagram

The above sequence diagram clearly shows how Aguka is able to report in a comprehensive manner with an aim of giving the Cooperative Manager insight on how the members of his cooperative do agriculturally. The Reporting Service is important in this case since it serves as the engine of data aggregation and representation in a chart form before presenting the information to the user. It should also be noted that there is an option for export both in PDF and Excel formats, something that shows how well the platform understands the needs of cooperative management in Rwanda's agriculture.

Activity Diagram

The activity diagram is a behavioral diagram used to depict actions or control flow in a system in the same way a flowchart or a data flow diagram does. The activity diagram depicts the control flow from the starting point to the end point by highlighting the different decision branches that exist during the process of performing the activity.

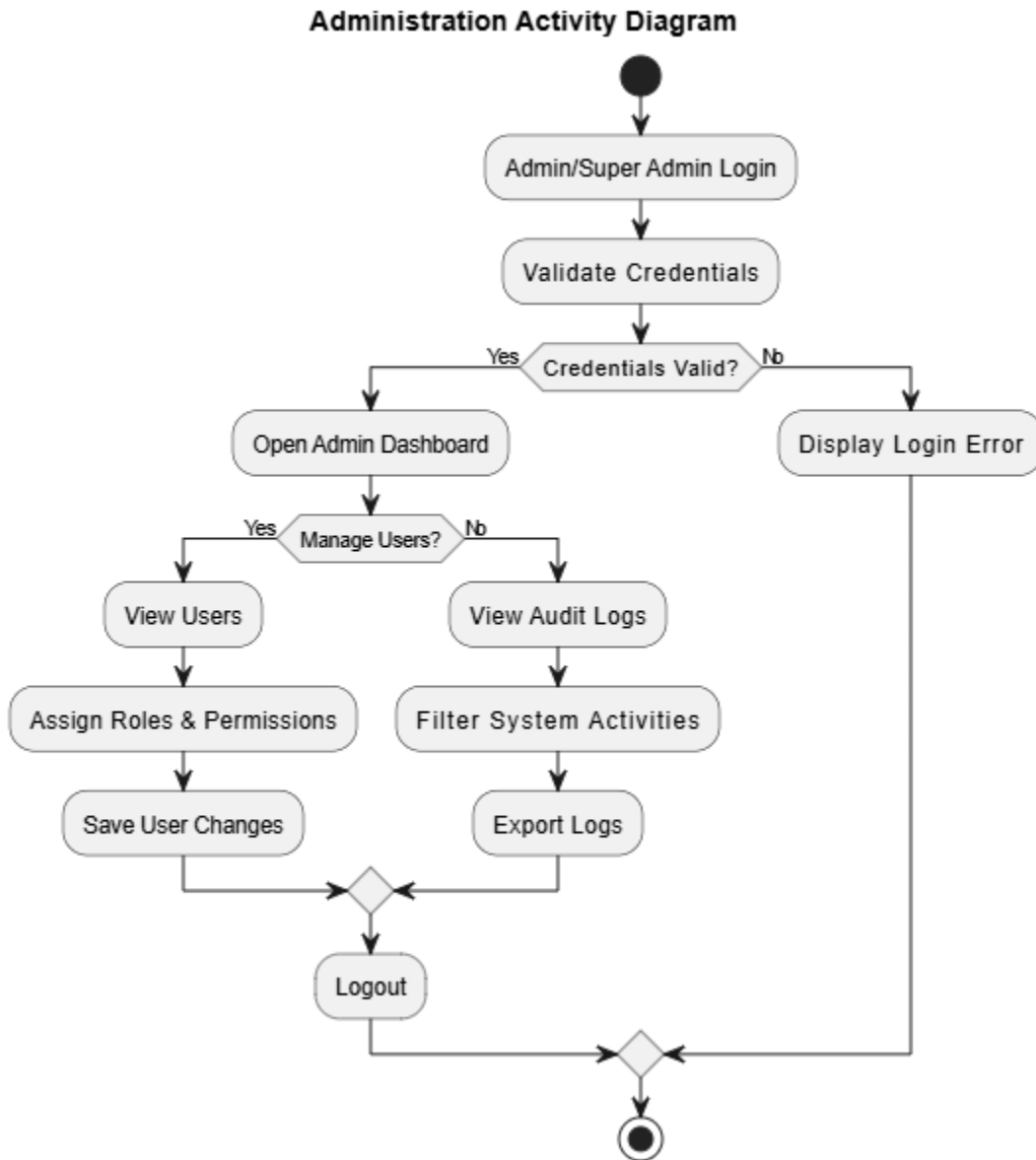


Figure 10: Administration Activity Diagram

The administration activity diagram illustrates the sequence of actions performed by Super Administrators and Administrators while managing the Aguka Smart Farming Kit. The process begins with user authentication, after which authorized administrators gain access to the administration dashboard. From the dashboard, they can perform various management functions such as managing users and access permissions, validating farm data, monitoring system activities, configuring notifications, generating reports, viewing audit logs, and managing backup and recovery operations. The activity flow ensures that administrative tasks are executed in a controlled and secure manner, thereby maintaining system integrity, data accuracy, and operational efficiency across the platform.

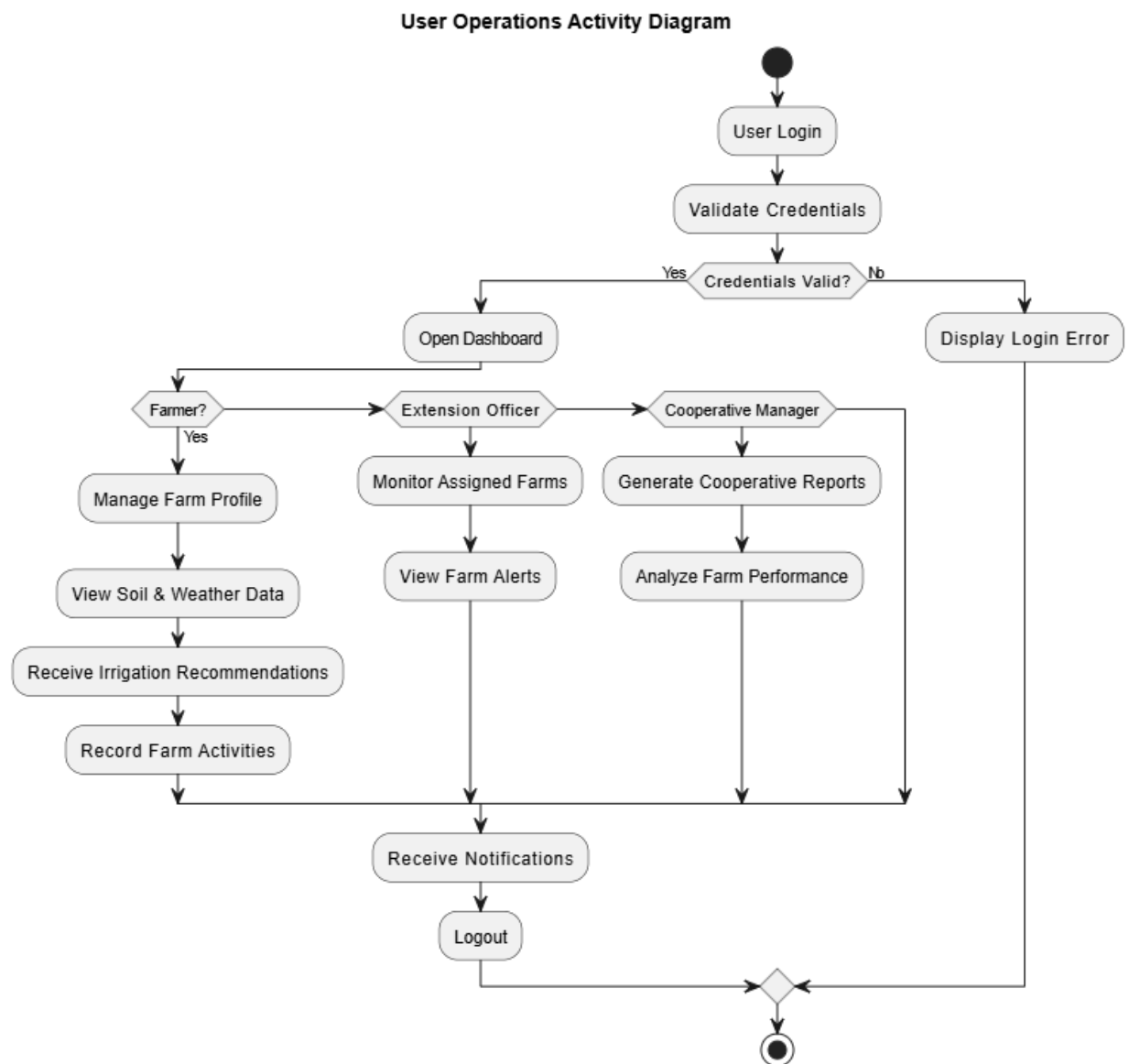


Figure 11: Activity Diagram for User operations

User Operations Activity Diagram depicts the process that involves operational users such as Farmers, Extension Officers and Cooperative Managers while working with Aguka Smart Farming Kit. The process begins with authentication and role verification which then leads to functionality access according to roles. Farmers are able to do farm management, soil & weather monitoring, farming activities management, get notifications and recommendations. Extension Officers can do farm monitoring, analysis of farm condition, give advisory services and make monitoring reports. Cooperative Managers are able to monitor member farmers, manage agricultural activities, get performance insights and make cooperative reports.

Database Diagrams

Database Designing is a combination of procedures that assist in the designing, developing, implementing and maintaining of enterprise data management systems. The well-designed database is easy to manage, increases data consistency and saves on disk storage costs. Database designer decides on how the data elements are correlated and data that will be stored. The main goals of database design are to develop logical and physical designs models for the proposed database system. The word database design can be used to refer to several things regarding design of an overall database system. Primarily, and more accurately, the database design can be viewed as the logical design of the base data structures for storing data. In relational database, these include tables and views. For the object databases, entities and relationships correspond to object classes and named relationships. But the term database design can also refer to overall process of design, not just base data structures, but also the forms and queries as well within the database management system.

Database diagram

The database schema is the framework that provides a logical representation of the entire database. It provides information about how the data is structured and the relations between them. The schema creates all the constraints that should be placed on the data.

The schema of the database includes all the entities of the database and their relationships. A database schema provides a detailed description of the database and can be represented through schema diagrams.

The database schema is designed by database designers to make programmers understand the database. (Chakraborty, 2021)

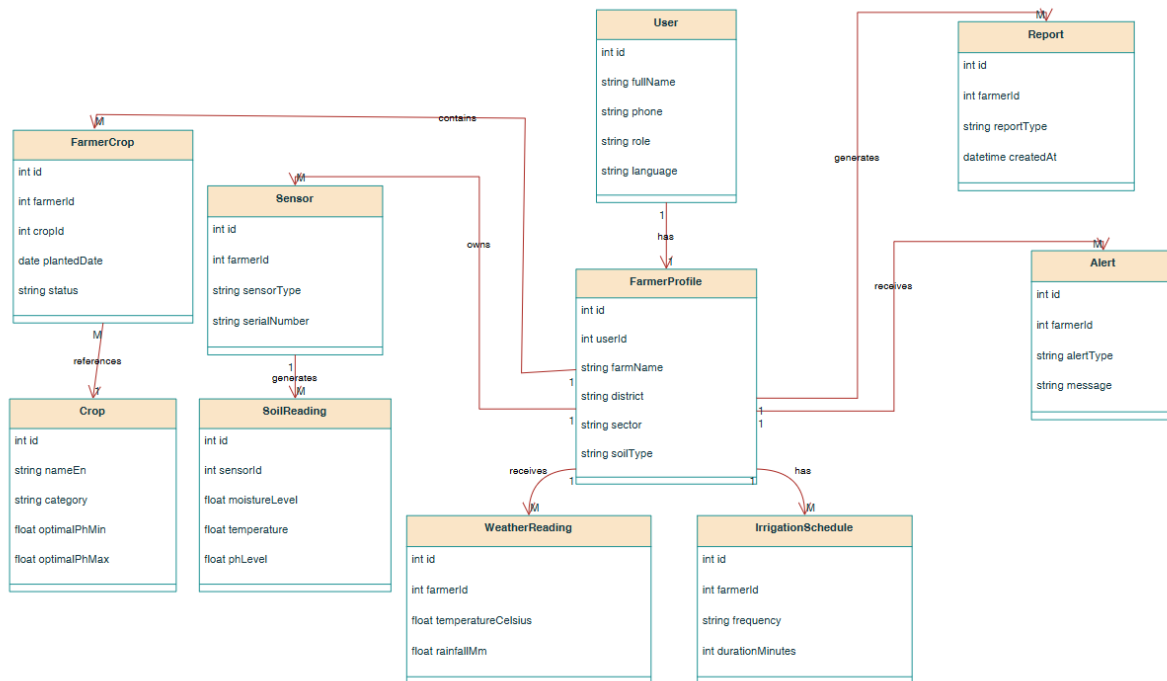


Figure 12: Diagram for Database Schema

Data Dictionary

Data Dictionary is a set of names, descriptions, and attributes of data elements being utilized or captured in a database, an information system, or a portion of a research project. Apart from providing interpretation, acceptable representations, and meaning, it defines the purpose and usage of data elements within the context of the system. It also provides detailed metadata about each data element, including its format, type, constraints, and relationships with other elements. This ensures consistency and standardization of data across the system, reducing ambiguity during development and usage. A well-structured data dictionary supports effective database design, system integration, and communication among developers and stakeholders. Data elements information is also offered by a data dictionary. Scope, characteristics, and guidance for using and applying data elements can be specified with the help of metadata in a data dictionary. (Wertz, 01 August 1989).

Table of Users

Column	Data Type	Constraint	Description
ID	UUID	PK, autoIncrement	ID for each user
fullName	VARCHAR (100)	NOT NULL	Full name of the user
Phone	VARCHAR (15)	UNIQUE, NOT NULL	User phone number
Email	VARCHAR (25)	UNIQUE	E-mail address of the user
PasswordHash	VARCHAR	NOT NULL	Password hash of the user
Role	ENUM	NOT NULL	Role of the user (Farmer, Admin, Officer)
Language	VARCHAR (25)	DEFAULT	Preferred language
isActive	BOOLEAN	DEFAULT TRUE	User account status
createAt	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Account creation date

Table 15: Users Table Data Dictionary

Farmer Profile Table

Field Name	Data Type	Constraints	Description
id	UUID	PK	ID for Farmer profile
userId	UUID	FK, NOT NULL	Foreign key to User table

FarmName	VARCHAR (100)		Name of the farm
district	VARCHAR (50)	NOT NULL	District where the farm is located
Sector	VARCHAR (50)	NOT NULL	Sector where the farm is located
soilType	VARCHAR (50)	NOT NULL	Type of soil on the farm
farmSizeHectares	DECIMAL	NOT NULL	Farm size in hectares

Table 16: Farmer Profile Table data dictionary

Crop Table

Field Name	Data Type	Constraints	Description
id	UUID	PK	Unique crop identifier
nameEn	VARCHAR (50)	NOT NULL	Crop name in English
Category	VARCHAR (50)	NOT NULL	Crop category
optimalPhMin	DECIMAL (4,2)		Minimum soil pH
optimalPhMax	DECIMAL (4,2)		Maximum soil pH
isActive	BOOLEAN	DEFAULT TRUE	Crop availability status

Table 17: Crop Table Data dictionary

FarmerCrop Table

Field Name	Data Type	Constraints	Description
------------	-----------	-------------	-------------

id	UUID	PK	Unique identifier for farmer crop record
farmerId	UUID	FK, NOT NULL	References FarmerProfile(id)
cropId	BIGINT	FK, NOT NULL	References Crop (Id)
plantedDate	TIMESTAMP	NOT NULL	Date when the crop was planted
status	VARCHAR (30)	DEFAULT 'planted'	Current crop status

Table 18: FarmerCrop Table data dictionary

Sensor Table

Field Name	Data Type	Constraints	Description
id	UUID	PK	Unique identifier for each sensor
farmerId	UUID	FK, NOT NULL	References FarmerProfile (id)
sensorType	ENUM	NOT NULL	Type of sensor
serialNumber	VARCHAR (100)	UNIQUE, NOT NULL	Sensor serial number
installationDate	TIMESTAMP		Date of sensor installation

Table 19: Sensor Table Data dictionary

SoilReading Table

Field Name	Data Type	Constraints	Description
------------	-----------	-------------	-------------

id	UUID	PK	Unique identifier for soil reading
sensorId	UUID	FK	References Sensor (id)
moisturePercent	NUMERIC (5,2)	NOT NULL	Soil moisture percentage
temperatureCelsius	NUMERIC (5,2)		Soil temperature in Celsius
phLevel	NUMERIC (4,2)		Soil pH level
readingAt	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Date and time of reading

Table 20: SoilReading Table Data Dictionary

WeatherReading Table

Field Name	Data Type	Constraints	Description
id	UUID	PK	Unique identifier for weather reading
farmerId	UUID	FK, NOT NULL	References FarmerProfile(id)
temperatureCelsius	NUMERIC (5,2)		Atmospheric temperature
humidityPercent	NUMERIC (5,2)		Humidity percentage
rainfallMn	NUMERIC (5,2)		Rainfall measurement in millimeters

readingAt	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Timestamp of weather reading
------------------	-----------	------------------------------	------------------------------

Table 21: WeatherReading Table Data Dictionary

IrrigationSchedule Table

Field Name	Data Type	Constraints	Description
id	UUID	PK	Unique identifier for Irrigation schedule
farmerId	UUID	FK, NOT NULL	References FarmerProfile (id)
frequency	VARCHAR (30)	NOT NULL	Irrigation frequency
durationMinutes	INTEGER		Irrigation duration in minutes
isActive	BOOLEAN	DEFAULT TRUE	Indicates if schedule is active

Table 22: IrrigationSchedule Table Data Dictionary

Alert Table

Field Name	Data Type	Constraints	Description
id	UUID	PK	Unique identifier for alert
farmerId	UUID	FK, NOT NULL	References FarmerProfile (id)
alertType	ENUM	NOT NULL	Type of alert generated
message	TEXT	NOT NULL	Alert message sent to farmer

createdAt	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Date and time alert was created
------------------	-----------	------------------------------	---------------------------------

Table 23: Alert Table Data Dictionary

AdvisoryReport Table

Field Name	Data Type	Constraints	Description
id	UUID	PK	Unique identifier
officeId	UUID	FK, NOT NULL	References Users(id) of Extension Officer
farmerId	UUID	FK, NOT NULL	References FarmerProfile(id) - the recipient
reportType	VARCHAR (50)	NOT NULL	'Pest_alert', 'irrigation_advice', 'fertilizer_recommendation'
title	VARCHAR (200)	NOT NULL	Report title
recommendations	TEXT	NOT NULL	The advisory content for the farmer
sentAt	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	When report was sent to farmer
status	VARCHAR (20)	DEFAULT 'draft'	Current report status ('draft', 'sent', 'read')

Table 24: Advisory Report Data Dictionary

CooperativeReport Table

Field Name	Data Type	Constraints	Description
id	UUID	PK	Unique identifier for cooperative report
cooperativeId	UUID	FK, NOT NULL	References the Cooperative the report belongs to
managerId	UUID	FK, NOT NULL	References Users(id) of the manager who generated it
reportType	TIMESTAMP (3)	VARCHAR (50)	Type of report (e.g., 'monthly_summary', 'performance_comparison')
startDate	TIMESTAMP	NOT NULL	Start of the reporting period
endDate	TIMESTAMP	NOT NULL	End of the reporting period
generatedAt	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	When the report was generated
fileUrl	TEXT		Path to generated PDF/Excel file
status	VARCHAR (20)	DEFAULT 'completed'	Status of report generation

Table 25: CooperativeReport Table Data Dictionary

SystemReport Table

Field Name	Data Type	Constraints	Description
id	UUID	PK	Unique identifier
adminId	UUID	FK, NOT NULL	References Users(id) of Admin
reportType	TIMESTAMP (3)	VARCHAR (50)	Type of report (<code>'financial_summary'</code> , <code>'user_activity'</code> , <code>'sensor_health'</code> , <code>'system_performance'</code>)
periodStart	TIMESTAMP	NOT NULL	Reporting period start
periodEnd	TIMESTAMP	NOT NULL	Reporting period end
fileUrl	TEXT		Path to generated PDF/Excel file
generatedAt	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Generation timestamp

Table 26: System Report Table Data Dictionary

System Architecture Design

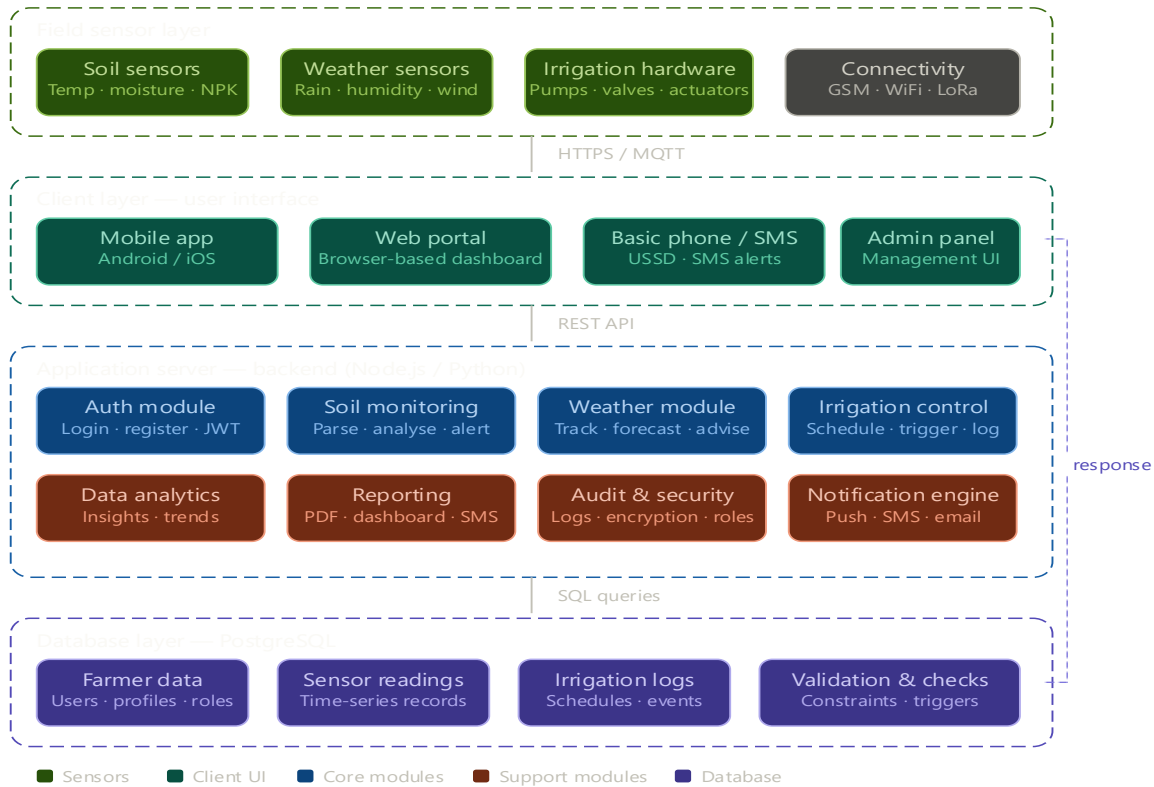


Figure 13: System Architecture Design

The System Architecture Design of the Aguka Smart Farming Kit presents the overall structure of the system and how its components interact to deliver smart farming services. It follows a multi-tier architecture composed of presentation, application, data, and external integration layers, ensuring scalability, security, and efficient communication between users, IoT systems, and backend services. The architecture is also explained using the MVC (Model-View-Controller) design pattern, which separates data handling, user interface, and control logic to improve system organization and development efficiency.

The system uses JWT-based authentication for secure login and role-based access control, while PostgreSQL with Prisma ORM manages data storage and operations. The frontend is developed using Flutter and JavaScript to provide responsive user interfaces, while backend services handle business logic, analytics, and system workflows.

CHAPTER 4

IMPLEMENTATION OF THE NEW SYSTEM

Introduction

This chapter explains the implementation of the “Aguka Smart Farming Kit.” It describes the technologies used to develop and run the application, as well as how the system was built and deployed. The chapter also includes screenshot demonstrations of the developed system to show how different features function in practice, along with the tests that were carried out to ensure system reliability and performance.

In addition, this chapter outlines the software and hardware compatibility requirements needed for the system to operate effectively. These requirements ensure that the application can run smoothly across different devices and environments, particularly in both mobile and web-based platforms used by farmers and agronomists.

Furthermore, the implementation process focuses on translating the system design into a working solution by integrating databases, user interfaces, and backend services. This ensures that all system components communicate properly and support the intended functionalities of the smart farming platform.

The development of the system followed a structured approach where each module was implemented separately before being integrated into the main platform. This modular approach helped in simplifying debugging and ensured that each component functioned correctly before full system integration. It also improved maintainability and allowed easier updates in the future.

During implementation, emphasis was also placed on ensuring system usability and performance efficiency. The user interface was designed to be simple and responsive to accommodate farmers with varying levels of technical experience. Additionally, optimization techniques were applied to ensure fast data processing and minimal system delays, especially when handling real-time agricultural data.

Technologies used

In order to create this web and mobile application, I used a variety of tools and technologies, including:

Frontend

Definition

Frontend development refers to the client-side part of the system that users interact with directly. It focuses on designing and implementing user interfaces that are responsive, interactive, and user-friendly.

Technology Stack

- **Flutter:** Used to develop the mobile application for Android devices with a modern and responsive user interface.
- **React.js:** Used for developing the web-based admin dashboard and management interfaces.
- **HTML (HyperText Markup Language):** Used for structuring web content.
- **CSS (Cascading Style Sheets):** Used for styling and designing the web interfaces.
- **JavaScript:** Used to add interactivity and dynamic functionality to the system.
- **Bootstrap/Tailwind CSS:** Used for responsive layouts and modern UI design.

Responsibilities of the Frontend

- Designing responsive and user-friendly interfaces for farmers, agronomists, suppliers, and administrators.
- Managing user interactions such as registration, login, onboarding, and dashboard navigation.
- Displaying real-time farming information and notifications.
- Ensuring compatibility across different screen sizes and devices.
- Improving user experience through modern design principles and accessibility features.

Frontend Tools

- **Visual Studio Code** for frontend coding and development.
- **Android Studio** for Flutter mobile application development and testing.
- **Figma** for UI/UX design and prototyping.
- **Chrome Developer Tools** for debugging and interface testing.
- **Git and GitHub** for version control and collaboration.

Backend

Definition

Backend development refers to the server-side part of the system responsible for managing application logic, databases, authentication, APIs, and communication between system components.

Technology Stack

- **Node.js:** Used as the server-side runtime environment.
- **Express.js:** Used to build RESTful APIs and manage backend services.
- **TypeScript:** Used to improve code maintainability, scalability, and type safety.
- **JSON Web Tokens (JWT):** Used for secure user authentication, session management, and role-based access control across all user roles
- **PostgreSQL:** Used for storing and managing structured system data.
- **Firebase Cloud Messaging (FCM):** Used for sending notifications and alerts to users.

AI Recommendation Engine

The Aguka Smart Farming Kit integrates a rule-based AI recommendation engine implemented as a backend service module. The engine analyses three data streams simultaneously:

Soil sensor readings moisture percentage, temperature, pH level, and NPK nutrient levels

Weather forecast data rainfall probability, temperature trends, and humidity levels

Historical crop activity data planting records, fertilizer logs, and growth stage information stored in PostgreSQL

Based on agronomic thresholds validated with IMBARAGA extension officers, the engine generates three types of recommendations:

1. **Irrigation Recommendations** When soil moisture falls below the crop-specific optimal threshold (for example, below 35% for maize), the system automatically generates an irrigation advisory: *"Soil moisture at 28% irrigation recommended within 24 hours. Estimated water needed: 15L/m²."*
2. **Fertilizer Alerts** When NPK nutrient readings fall below the optimal levels for the registered crop type and growth stage, the system generates: *"Nitrogen levels low for maize at growth stage 3 apply urea fertilizer within 7 days."*
3. **Pest and Disease Risk Warnings** When temperature and humidity combinations match historical pest outbreak conditions, the system generates: *"High humidity (87%) and temperature (26°C) conditions match Fall Armyworm outbreak pattern inspect farms in Gakenke district."*

These recommendations are automatically delivered to farmers via push notifications through Firebase Cloud Messaging and displayed on their dashboards in real time.

Responsibilities of the Backend

- Managing user authentication and authorization.
- Processing and storing farming data securely.
- Handling role-based onboarding for farmers, agronomists, suppliers, and administrators.
- Providing APIs for communication between mobile applications and the database.
- Managing notifications, reports, and smart farming services.
- Ensuring data security, scalability, and system performance.

Backend Tools

- **Visual Studio Code** for backend development.
- **Postman** for API testing and validation.

- **PostgreSQL** for database management and authentication.
- **Git and GitHub** for source code management and collaboration.

IoT Sensor Simulation

The Aguka Smart Farming Kit was designed to integrate with physical IoT soil and weather sensors deployed on farms. However, due to the high cost of procuring and deploying physical sensor hardware across multiple smallholder farms in Rwanda, the sensor data in this project is generated through a software-based simulation module developed as part of the system backend.

The simulation module generates realistic sensor readings including soil moisture percentage, soil temperature, pH levels, NPK nutrient values, rainfall measurements, humidity levels, and ambient temperature. These values are produced based on seasonal agricultural patterns and crop growth stages typical of Rwanda's Northern and Southern Provinces, where IMBARAGA member farmers are concentrated.

This approach is consistent with widely accepted practice in smart farming research and system prototyping. Several published studies in precision agriculture follow the same methodology of simulating sensor data to validate system logic and AI recommendation engines before committing to hardware investment. The simulation enabled full end-to-end testing of all system modules including the AI recommendation engine, irrigation scheduling, push notification triggers, and Extension Officer monitoring dashboards without any hardware dependency.

The simulation is implemented as a dedicated service module (`simulation.service.ts`) in the Node.js backend. It can be replaced with real sensor API calls in a future production deployment without any changes to the AI engine or frontend interfaces, since the data format and API endpoints remain identical.

Physical IoT sensor deployment is identified as the primary recommendation for future enhancement of the Aguka Smart Farming Kit, as described in Chapter 5 Conclusions and Recommendations.

Presentation of the New System

This section shows the results of implementing this system. It includes user interfaces, web pages, and mobile application screens generated through the system's code execution. Some interfaces are displayed as screenshots, while others are explained in detail in the following subsections.

Welcome Page

Description:

The welcome page serves as the entry point to the Aguka ecosystem. It features a clean, nature-inspired aesthetic with a prominent hero section titled "Grow more. Worry less." This immediately communicates the value proposition of the platform.

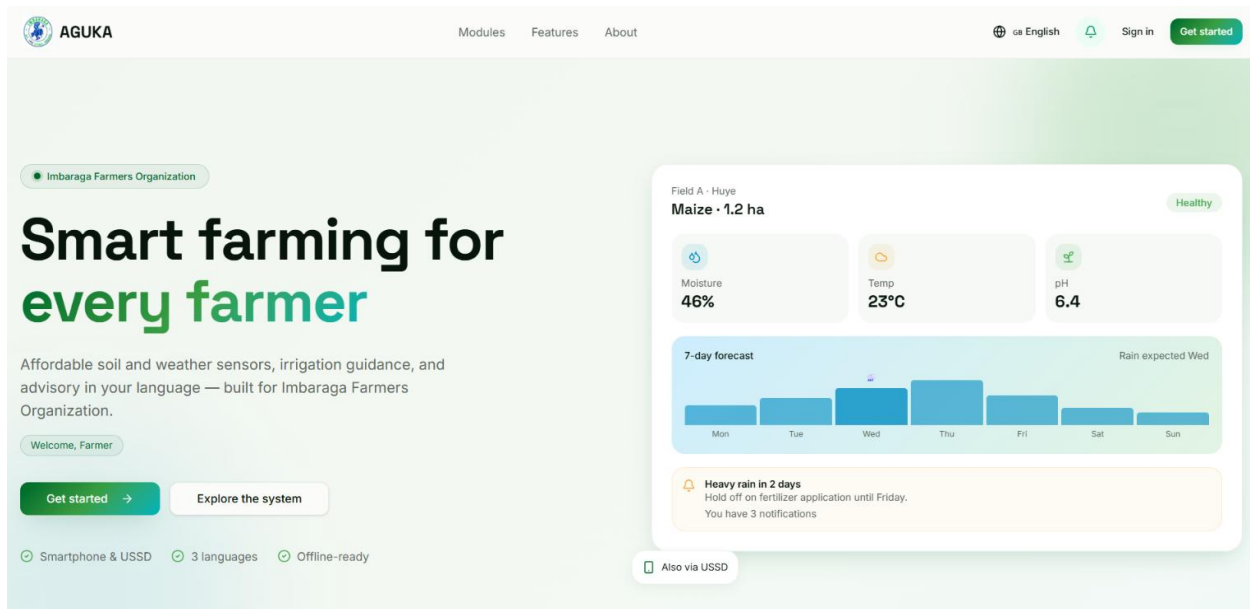


Figure 14: Welcome Page

Sign up Page

Description:

The registration process consists of two mobile-friendly screens that guide users from identity collection to security setup. The interface is clean, clearly indicates progress ("STEP 1 OF 2" → "STEP 2 OF 2"), and introduces the system's value proposition throughout.

Step 1: Personal Information

This first screen gathers essential identity details while establishing trust through impact metrics (20 active farmers, 10 cooperatives, 10 districts). Users provide their Full Name, optional Email Address, and select their System Role from three options: Farmer, Extension Officer, or Cooperative Manager.

Step 2: Security Details

The second screen focuses on authentication credentials. Recognizing that many farmers have mobile phones but not necessarily email addresses, the system uses the Rwandan phone number format (+250) as the primary username/identifier. Users provide their Phone Number, create a Password (minimum 8 characters), confirm it, and agree to the Terms of Service and Privacy Policy before completing signup.

The screenshot displays the registration interface for the AGUKA Smart Farming Kit. The left sidebar features the AGUKA logo and the title 'Smart Farming Kit for Smallholder Farmers'. Below the title, it describes the kit's capabilities: connecting soil sensors, monitoring hyper-local weather, implementing automated irrigation, and syncing with agricultural cooperatives. Three impact metrics are listed: 20 active smallholder farmers, 10 cooperatives connected, and 11 districts covered. The main content area on the right is titled 'Create Account' and includes a progress indicator for 'STEP 1 OF 2' and 'PERSONAL INFO'. The form fields include 'Full Name' (with an example 'e.g. Claudette Uwimana'), 'Email Address (Optional)' (with an example 'claudette@example.com'), and 'Assign System Profile Role' with three options: 'FARMER', 'EXTENSION OFFICER', and 'COOPERATIVE MANAGER'. A green 'Continue' button with a right arrow is positioned below the role selection. At the bottom, a link for 'Already have an account? Sign in' is provided.

Figure 15: Sign up page

Login Page

Description:

The login page serves as the authentication gateway for returning users. It maintains consistent branding with the registration flow while offering a streamlined, focused interface for quick access. The page recognizes that phone number (+250 format) is the primary identifier for farmers, reinforcing the system's mobile-first approach for users who may not have email addresses.

Figure 16: Login page

The page includes a “Start” button at the bottom, which serves as the primary call-to-action. Users tap this button to proceed to the authentication or registration workflow, allowing them to log in or sign up to access the system’s features such as submitting monitoring data, reporting incidents, and viewing interactive maps.

Farmer Dashboard

Description:

This Dashboard is the primary interface that farmers see after logging into the Aguka Smart Farming Kit system. It presents a comprehensive, at-a-glance overview of the farmer's registered farm(s), including key environmental data, crop information, and weather conditions.

In cases where the farmer has no farm registered yet, the dashboard displays the core interface layout with placeholder or empty data, signaling that the farmer needs to set up their farm profile before full functionality becomes available. After successful registration, the farmer is redirected to the dashboard, where they can begin managing their farm. It is important to note that the system uses simulated IoT device data for demonstration and testing purposes, allowing farmers to interact with realistic sensor readings even when physical sensors are not yet deployed.

The dashboard prioritizes critical information at a glance while providing intuitive navigation to deeper system features such as recommendations, irrigation management, and community discussions. It serves as the central hub for farmers to monitor farm health, make informed decisions, and receive actionable insights.

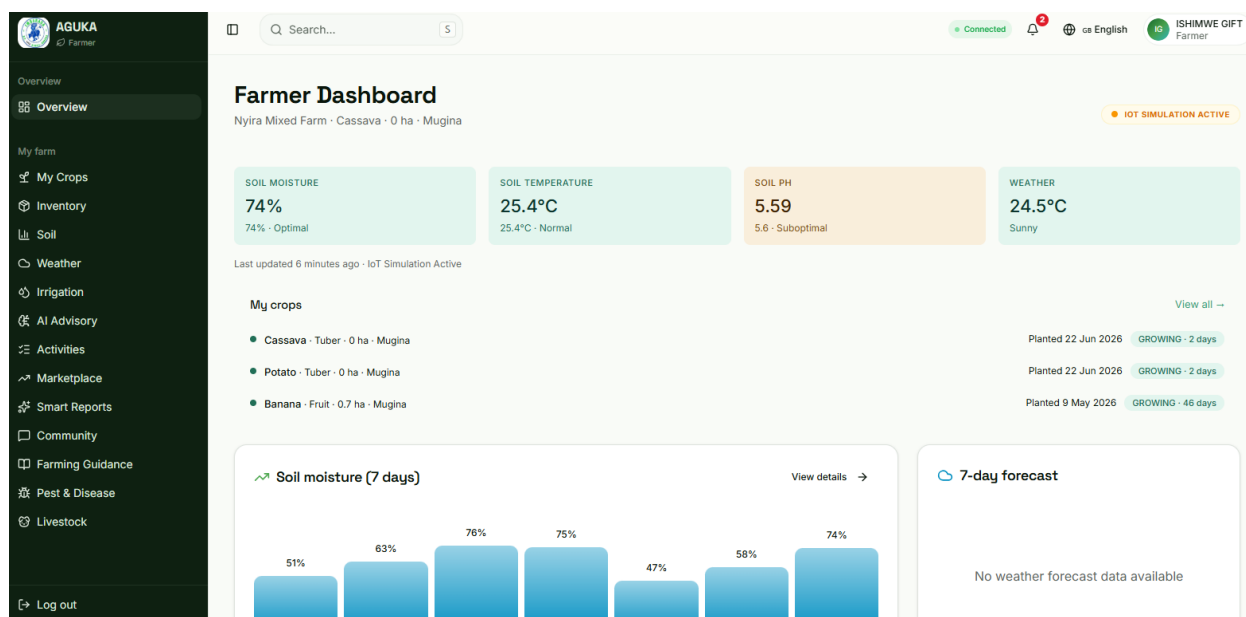


Figure 17: Farmer dashboard

Field Officer Dashboard

Description:

The Extension Officer, also referred to as the Field Officer, is a supervisory user role within the Aguka Smart Farming Kit, specifically designed for agricultural extension officers operating within organizations such as Imbaraga Farmers Organization.

Unlike the farmer-facing dashboard, the Extension Officer's interface is data-aggregated and risk-focused. It provides a high-level operational view of all assigned farmers and farms under the officer's supervision. The dashboard is built to prioritize early warning signals including pest risks and weather alerts—and to highlight action items that require immediate officer intervention.

The dashboard displays key metrics such as number of assigned farmers, farms monitored, pending visits, and active risks, enabling the officer to respond quickly and effectively to emerging threats and operational needs.

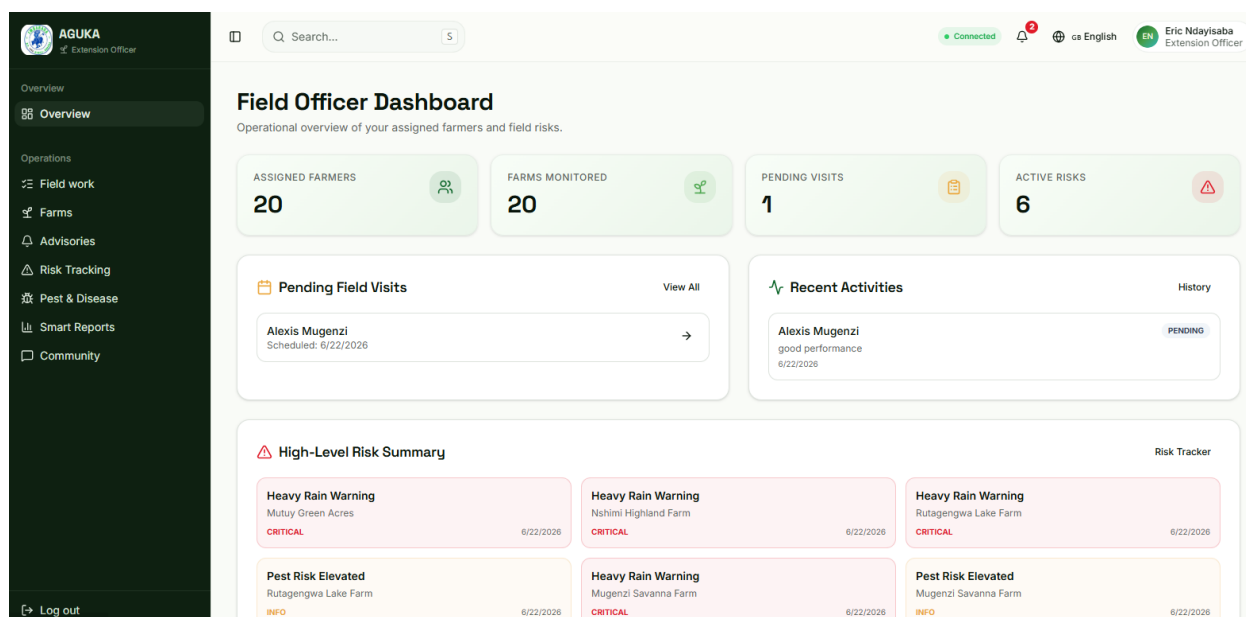


Figure 18: Field Officer Dashboard

Cooperative Manager Dashboard

Description:

The Cooperative Manager Dashboard is designed for leadership roles within Imbaraga Farmers Organization who oversee multiple farmers, farm plots, and shared assets across the cooperative. This interface emphasizes aggregate analytics, member performance comparisons, and cooperative-level resource management.

The dashboard provides a high-level command centre view, enabling cooperative managers to monitor the overall health and productivity of the network. It prioritizes key performance indicators, resource distribution, and member rankings, allowing managers to identify top performers, track cooperative-wide trends, and make data-driven decisions that benefit the entire organization.

This dashboard enables cooperative managers to supervise farmer performance, allocate resources efficiently, track member dues and expenses, and generate cooperative-level reports to support strategic planning and advisory efforts.

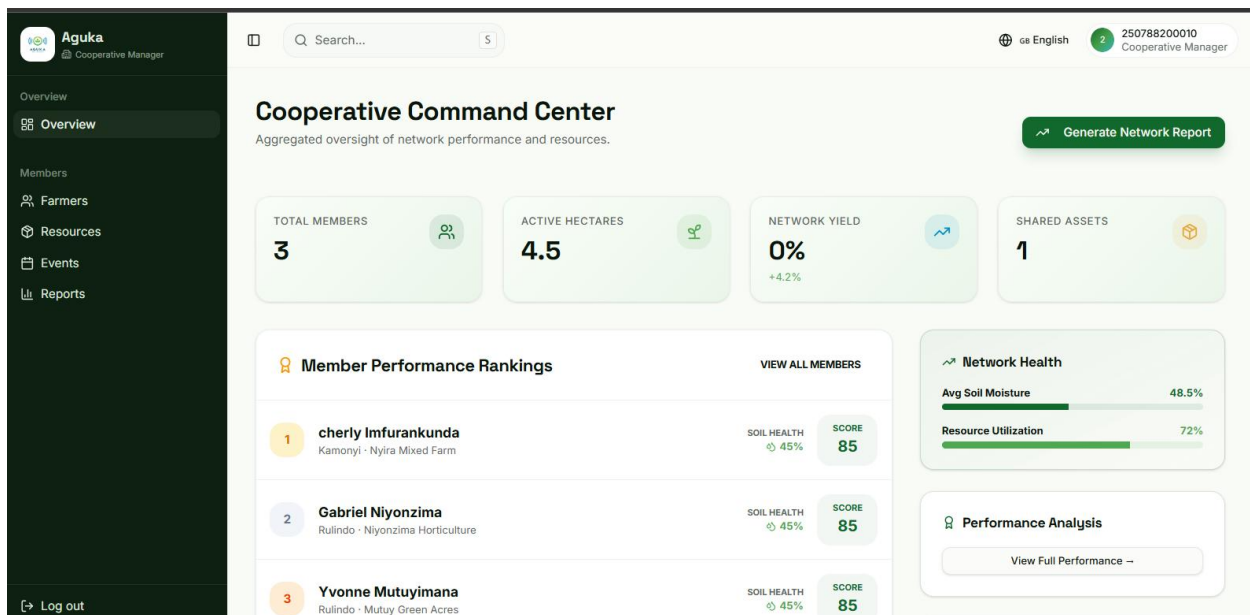


Figure 19: Cooperative Manager Dashboard

Admin Overview Dashboard

Description:

The Admin Overview Dashboard is the highest level of access within the Aguka Smart Farming Kit system, designed for platform administrators who manage the entire ecosystem. This interface focuses on system-wide aggregates, sensor data integrity monitoring, and user validation workflows.

Unlike the Farmer, Field Officer, or Cooperative Manager dashboards which focus primarily on agricultural operations this dashboard emphasizes platform governance, data quality assurance, and overall system health. It provides administrators with a centralized view of system usage, farm activity, and pending data validations, enabling them to ensure the platform runs smoothly and reliably.

This dashboard enables administrators to validate farmer data, monitor system performance, track sensor connectivity, and manage user access across the entire Aguka platform, ensuring data integrity and operational stability.

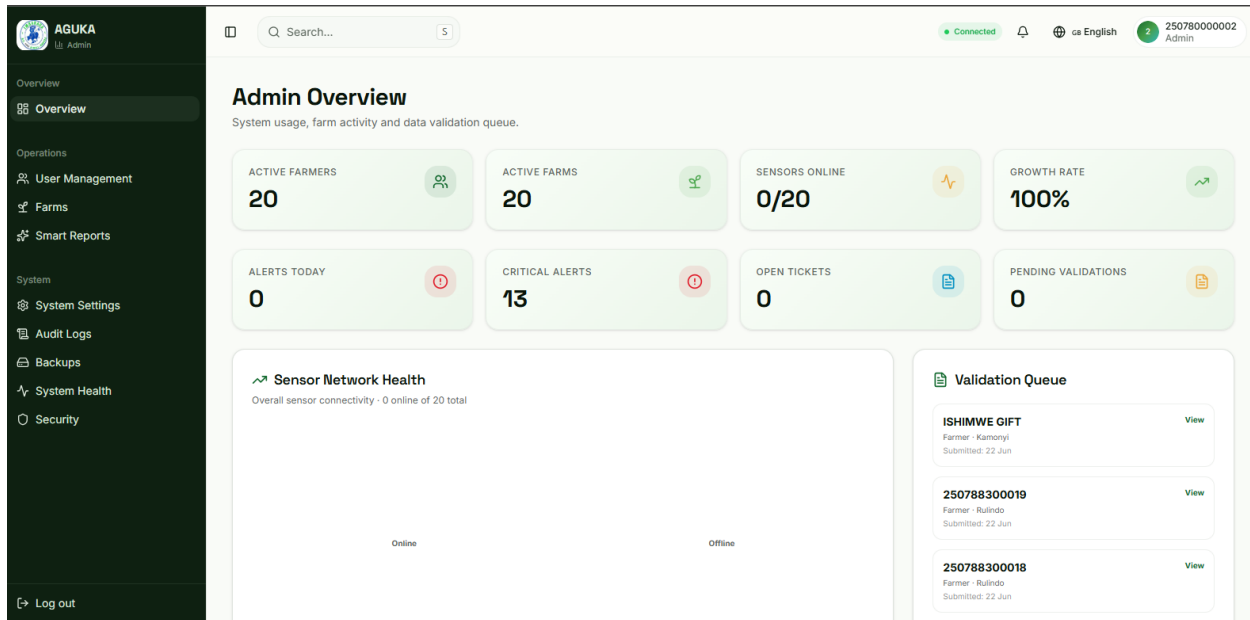


Figure 20: Admin Overview Dashboard

Super Admin Console Overview

Description:

The Super Admin Console is the most privileged view in the Aguka Smart Farming Kit, granting unrestricted control over all modules, users, settings, and infrastructure. Unlike the standard Admin dashboard which focuses primarily on user and farm management this interface emphasizes infrastructure health, system-wide metrics, and oversight of all administrative actions.

The Super Admin Console provides a comprehensive command centre for platform governance, enabling super administrators to monitor system performance, manage user roles, review audit trails, and configure core system settings. It is designed for individuals responsible for the overall stability, security, and scalability of the system

This console enables super administrators to manage users and access control, configure system settings, view audit logs and monitor activity, manage backups and recovery, and generate system and financial reports ensuring the entire Aguka ecosystem operates securely, efficiently, and reliably.

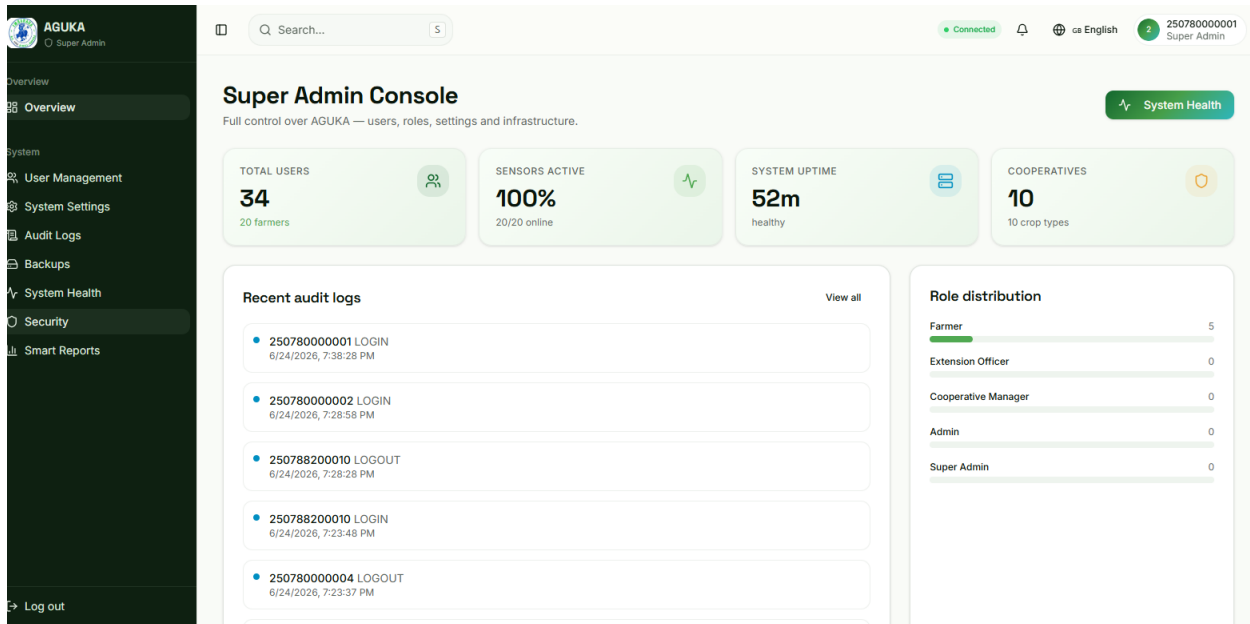


Figure 21: Super Admin Console Overview

Super Admin Report

Description:

The Super Admin Reports section is a dedicated reporting module within the Super Admin Console of the Aguka Smart Farming Kit system. This interface provides verifiable system reports with real-time data, enabling super administrators to access cryptographically secured, interactive, and auditable reports across all platform operations.

Unlike standard reporting tools, this module emphasizes data integrity, transparency, and comprehensive oversight, making it essential for platform governance, compliance, and strategic decision-making. Each report includes interactive charts, detailed metrics, and verifiable data logs that can be audited for accuracy and authenticity.

Key reports available in this section include:

- Audit Log Report
- System Health Report
- Backup & Recovery Report

- Security & Threat
- National Performance Report

Each report includes an **Open Report** button to view full details, explore interactive visualizations, and access verifiable data logs for comprehensive analysis and audit purposes.

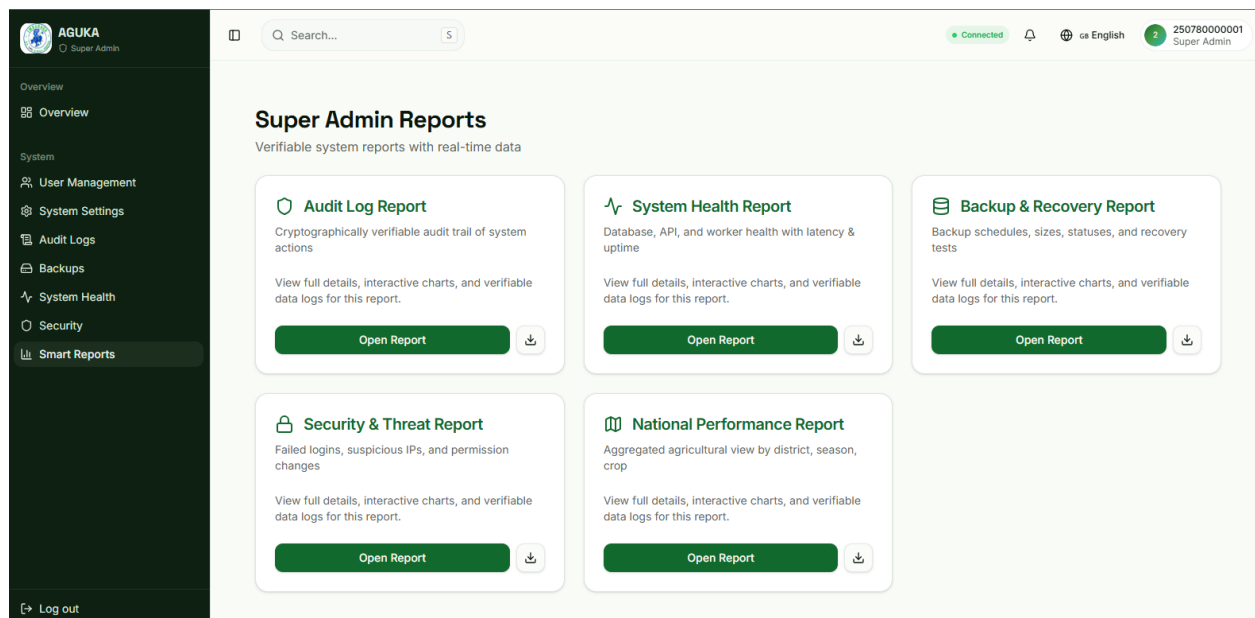


Figure 22: Super Admin Report

Audit Log Report (PDF)

Description

The Audit Log Report is a cryptographically verified PDF export from the Aguka Smart Farming Kit system, designed for super administrators and auditors. It provides a comprehensive record of all system user activities, including logins, logouts, and key actions, with a focus on data integrity and transparency.

The report is confidential, timestamped, and includes the generating user's ID and role for full traceability.



SUMMARY OVERVIEW

TOTAL LOGS

55

ACTIVE USERS

6

SUCCESS RATE

100.0%

REPORTING PERIOD

25 May 2026 - 24 Jun 2026

TOP USERS BY ACTIVITY

USER EMAIL	TOTAL ACTIONS
alice.nyira@aguka.rw	19
superadmin@aguka.rw	13
officer1@aguka.rw	9
manager.rulindo@aguka.rw	5
officer2@aguka.rw	5

RECENT SYSTEM LOGS

TIME (UTC)	USER	ACTION	STATUS
24 Jun 2026 17:38 UTC	superadmin@aguka.rw	LOGIN	SUCCESS
24 Jun 2026 17:28 UTC	admin@aguka.rw	LOGIN	SUCCESS
24 Jun 2026 17:28 UTC	manager.rulindo@aguka.rw	LOGOUT	SUCCESS
24 Jun 2026 17:23 UTC	manager.rulindo@aguka.rw	LOGIN	SUCCESS
24 Jun 2026 17:23 UTC	officer2@aguka.rw	LOGOUT	SUCCESS
24 Jun 2026 17:09 UTC	officer2@aguka.rw	LOGIN	SUCCESS
24 Jun 2026 17:09 UTC	alice.nyira@aguka.rw	LOGOUT	SUCCESS



24 Jun 2026 17:08 UTC	alice.nyira@aguka.rw	LOGIN	SUCCESS
24 Jun 2026 17:07 UTC	superadmin@aguka.rw	LOGIN	SUCCESS
23 Jun 2026 16:16 UTC	superadmin@aguka.rw	LOGIN	SUCCESS

DOCUMENT INTEGRITY HASH (SHA-256)
0e51e5b7f7f35bd8ff32de45818fded9a0875ec0f8c5478148ea894eba943b168

Figure 23: PDF Report

Software Testing

Software testing plays an important role in ensuring the quality, reliability, and efficiency of the **Aguka Smart Farming Kit**. Testing was conducted to verify whether the system functions according to the specified requirements and effectively supports farmers, cooperative managers, extension officers, administrators, and super administrators.

The testing process helped identify and correct errors before deployment, ensuring the system operates correctly and provides a smooth user experience.

When conducting software testing for the Aguka Smart Farming Kit System, the following questions were considered:

Does the system comply with the requirements and standards used during development?

Does the system function correctly according to user expectations?

Can the system be implemented effectively to support smart farming activities and management operations?

These considerations ensured that the developed system met both functional and technical requirements.

Types of Software Testing

Unit Testing

Unit testing was performed to verify the functionality of individual modules and components of the Aguka Smart Farming Kit System. Each module was tested independently to ensure it operates correctly according to its specifications.

The following modules were tested:

- User authentication and login module
- Farmer registration module
- Role-based onboarding module
- Farm verification module
- Weather monitoring module
- Soil monitoring module

- Report generation module

Unit testing helped identify logical errors and improve the reliability of individual system components before integration.

Integration Testing

Integration testing was conducted to ensure proper communication and interaction between system components.

The following integrations were tested:

- Frontend communication with backend APIs
- Integration between React dashboard and Node.js backend
- Flutter mobile application integration with backend services
- Database integration with PostgreSQL and Firebase Cloud Messaging services

This testing ensured that all modules work together correctly and that data flows properly throughout the system.

System Testing

System testing was performed on the complete Aguka Smart Farming Kit System after integrating all modules.

The purpose of this testing was to verify:

- Overall system functionality
- User interface responsiveness
- Dashboard operations
- Role-based access control

The entire application was tested under different scenarios to ensure that all requirements were successfully achieved.

The testing focused on evaluating whether the system meets user needs and supports agricultural activities effectively. Feedback from users helped improve usability and overall system experience.

Hardware and Software Compatibility Requirements

Client-Side Requirements

Operating Systems

- Windows 10 or later
- Android 8.0 or later

Web Browsers

- Google Chrome
- Mozilla Firefox
- Microsoft Edge

Mobile Platform

- Android smartphones and tablets

Frontend Technologies

- React.js
- Flutter
- HTML, CSS, JavaScript

Processor Requirements

- Minimum dual-core processor
- Quad-core processor recommended for better performance

RAM Requirements

- Minimum 4 GB RAM
- 8 GB RAM recommended

Mobile RAM

- Minimum 2 GB RAM
- 4 GB RAM recommended

Storage Requirements

- Minimum 256 GB HDD/SSD
- SSD recommended for faster performance

Display Requirements

- Minimum resolution: 1280×720
- Recommended: 1920×1080 or higher

Network Requirements

- Stable internet connection
- Minimum 5 Mbps download speed
- 4G mobile internet recommended for mobile users

Peripherals

- Keyboard and mouse
- Smartphone touchscreen
- Speakers and microphone for notifications and communication

Server-Side Requirements

Backend Framework

- Node.js with Express.js
- TypeScript support

Database

- PostgreSQL Database Server
- Firebase services integration

Authentication Services

- Firebase Authentication

Notification Services

- Firebase Cloud Messaging (FCM)

Frontend Deployment

- Vercel hosting platform

Backend Hosting

- Cloud server hosting with secure configuration

Version Control

- Git and GitHub repositories

Server Processor

- Multi-core processor recommended

Server RAM

- Minimum 4 GB RAM
- 8 GB or more recommended

Server Storage

- Minimum 256 GB SSD storage

Network Requirements

- Reliable high-speed internet connection

CHAPTER 5

CONCLUSIONS AND RECOMMENDATIONS

Conclusion

The Aguka Smart Farming Kit represents an innovative solution developed to address challenges faced by smallholder farmers in Rwanda through the integration of modern agricultural technologies and digital management systems. The project was designed to improve smart farming practices by providing farmers, cooperative managers, extension officers, and administrators with a centralized platform for farm management, monitoring, and agricultural decision-making.

Through the implementation of the system, farmers are able to access smart farming services such as weather monitoring, irrigation guidance, farm activity tracking, and agricultural advisories using mobile and web technologies. The system enhances communication and collaboration among stakeholders while improving the efficiency of agricultural operations.

The integration of real-time data management, role-based dashboards, and multilingual support ensures accessibility and usability for different categories of users, especially local farmers. By supporting English, Kinyarwanda, and French languages, the system promotes inclusiveness and wider adoption within farming communities.

The project also contributes significantly to digital transformation in agriculture by enabling better farm data management, verification processes, and reporting mechanisms. Through technologies such as React.js, Flutter, Node.js, Express.js, and PostgreSQL, the system provides a secure, scalable, and modern agricultural platform.

Ultimately, the Aguka Smart Farming Kit System is more than just a farming application; it is a smart agriculture initiative that supports digital agriculture transformation, promotes sustainable farming practices, improves productivity, and strengthens agricultural management within Rwanda. The project contributes toward improving food security, farmer empowerment, and sustainable agricultural development.

Recommendations

To Farmers and Cooperatives

It is recommended that farmers and agricultural cooperatives adopt and actively use the Aguka Smart Farming Kit System to improve farm management and agricultural productivity. By utilizing features such as weather monitoring, irrigation guidance, crop tracking, and farming advisories, farmers can make informed decisions that enhance crop production and reduce farming risks.

Farmers are also encouraged to regularly update farm data within the system to improve monitoring accuracy and support better agricultural planning.

To Agricultural Institutions and Government Authorities

It is recommended that agricultural institutions and government agencies support the implementation and expansion of the Aguka Smart Farming Kit System across different farming communities in Rwanda. Integration of the system into national agricultural programs can improve agricultural data management, farmer support services, and smart farming initiatives.

Government institutions may also collaborate with developers and agricultural stakeholders to enhance digital agriculture policies and encourage technology adoption among farmers.

To AUCA

It is recommended that the Adventist University of Central Africa (AUCA) continue supporting research and innovation in smart agriculture and digital transformation projects. The university should encourage students and researchers to further improve the Aguka Smart Farming Kit System by integrating advanced technologies such as Artificial Intelligence (AI), IoT sensors, predictive analytics, and machine learning for smarter agricultural decision-making.

AUCA can also use the project as a learning platform for practical software engineering, information technology, and agricultural innovation research.

To Future Researchers

Future researchers are encouraged to continue improving the Aguka Smart Farming Kit System by conducting additional studies on smart farming technologies and sustainable agriculture solutions.

Researchers may focus on:

- Integration of IoT-based soil and weather sensors
- Artificial Intelligence for crop disease prediction
- Smart irrigation automation
- Offline mobile functionality for rural areas
- Big data analytics for agricultural forecasting
- Integration with national agricultural databases and services

Researchers should also work closely with farmers, cooperatives, and agricultural institutions to ensure that future improvements align with real agricultural challenges and user needs.

REFERENCES

Books

- Creswell, J. W. (2014). Research Design: Qualitative, Quantitative, and Mixed Methods Approaches (4th ed.). Thousand Oaks, CA: SAGE Publications.
- Kothari, C. R. (2012). Research Methodology: Methods and Techniques (2nd ed.). New Delhi: New Age International Publishers.

Journals

- Ahirav, D. (2024). Part 6: Introduction to Express.js – Simplifying Web Server Development in Node.js.
- Anupindi, R., & Others. (2018). E-Procurement and Inventory Performance: Evidence from Panel Data on the Internet Integration in Procurement Processes.
- Lim, E. (2021). WebSockets and Real-Time Applications. Medium.

Websites

- Biofin. (2021). Protecting Nature Through Digital Innovations in Rwanda. Retrieved from <https://www.biofin.org/news-and-media/protecting-nature-through-digital-innovations-rwanda>
- Dan, A., & Rachel, N. (2023). Introducing React.dev. Retrieved from <https://react.dev/blog/2023/03/16/introducing-react-dev>
- FAO. (2024). Food and Agriculture Organization of the United Nations in Rwanda. Retrieved from <https://www.fao.org/rwanda/en/>
- Flutter. (2024). Flutter FAQs. Retrieved from <https://docs.flutter.dev/resources/faq>
- Flutter. (n.d.). Architecture Overview. Retrieved from <https://docs.flutter.dev/resources/architectural-overview>
- GeeksforGeeks. (2024). Introduction to Java. Retrieved from <https://www.geeksforgeeks.org/introduction-to-java/>

GeeksforGeeks. (2023). Apache Maven. Retrieved from <https://www.geeksforgeeks.org/apache-maven/>

GeeksAnts. (2023). Top 5 Flutter UI Libraries. Retrieved from <https://geekyants.com/en-us/blog/top-5-flutter-ui-libraries-for-2023>

GovStack Global. (2024). Modernising Tree Management: Improving Rwanda's Forests Through Innovative Collaboration. Retrieved from <https://www.govstack.global>

Lucidchart. (2024). What is Unified Modeling Language (UML)? Retrieved from <https://www.lucidchart.com/pages/what-is-UML-unified-modeling-language>

MDN Web Docs. (2024). CSS: Cascading Style Sheets. Retrieved from <https://developer.mozilla.org/en-US/docs/Web/CSS>

Medium. (2023). What is Node.js and How Does It Work? Retrieved from <https://medium.com/@asiandigitalhub/what-is-node-js-and-how-it-work-490f5ecba665>

MUI Core. (2024). Material UI Overview. Retrieved from <https://mui.com/material-ui/getting-started/>

e-Ihuriro. (2019). Environment and Climate Change Analysis Rwanda. Retrieved from <https://e-ihuriro.rcsprwanda.org>

Ministry of Environment Rwanda. (2020). How Rwanda is Putting its Forests First in the Fight Against Climate Change. Retrieved from <https://www.environment.gov.rw>

JetBrains. (2024). IntelliJ IDEA Overview. Retrieved from <https://www.jetbrains.com/help/idea/discover-intellij-idea.html>

APPENDICES

Curriculum Vitae

Adolphe Uwayo

IT Specialist | Software & Database Developer

 Kigali, Rwanda

 +25078188846

Summary

Motivated and detail-oriented Information Technology student specializing in Information Management at Adventist University of Central Africa (AUCA). Experienced in system development, database design, and digital innovation.

Education

Adventist University of Central Africa (AUCA)

B.Sc. in Information Technology (Major: Information Management)

Expected Graduation: 2026

Skills

- **Programming:** Java, PHP, C, C++, Python, JavaScript, HTML, CSS
- **Databases:** MySQL, PL/SQL, PostgreSQL
- **Frameworks & Tools:** React.js, Node.js, Figma, WordPress
- **IT Support:** Troubleshooting, Networking, Server Management

Certifications

- **Oracle Database Design (PL/SQL)**
- **Network Operations**
- **Data Analysis**

Languages

- **English** (Fluent)
- **Kinyarwanda** (Fluent)
- **French** (Basic)

Data Collection Letter



Adventist University of Central Africa

P.O. Box 2461 Kigali, Rwanda | www.auca.ac.rw | Info@auca.ac.rw

Faculty of Information Technology

March 17, 2026

TO: CEO

IMBARAGA FARMERS ORGANIZATION

Dear Sir/Madam,

Re: Request for Data Collection

The bearer, **UWAYO Adolphe** is a student of Adventist University of Central Africa in the Faculty of Information Technology, major in Information Management Department and he is working on a dissertation entitled: "AGUKA SMART FARMING KIT"

The purpose of this letter is to request for your permission to allow him to collect data in your organization.

Any assistance given to him will be highly appreciated.

Yours sincerely;

A handwritten signature in blue ink, appearing to read 'Mr. Ishimwe Prince', is written over a circular blue stamp. The stamp contains the university's logo and the text 'ADVENTIST UNIVERSITY OF CENTRAL AFRICA' and 'RWANDA'.

Mr. ISHIMWE Prince
HOD||Software Engineering
Tel: +250 782 974 477
Email: prince.ishimwe@auca.ac.rw

Approval Letter from Organization



Umurvango w'Abahinzi n'Aborozi bo mu Rwanda « IMBARAGA » Organisation des
Agriculteurs et Eleveurs du Rwanda
BP 1462, Tél 0788610584, 0788843303 ; E mail info@imbaraga.org; www. imbaraga.org

November 3, 2025

Uwayo Adolphe
Student of Adventist University of Central Africa,

Subject: Approval of your request

We are pleased to inform you that your request to conduct your research project at Imbaraga Farmers' Organization has been approved. You are now authorized to begin your research activities while respecting the organization's guidelines and maintaining professionalism throughout your work.

We wish you great success in your research!

Ignace Nteziryayo

Imbaraga North Coordinator

